

DEVELOPMENT OF RAPID SMART HYDROGEL SENSORS
WITH RESISTIVE CHANNEL SENSING FOR
POINT-OF-CARE APPLICATIONS

by

Hsuan-Yu Leu

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STATEMENT OF DISSERTATION APPROVAL

The dissertation of Hsuan-Yu Leu

has been approved by the following supervisory committee members:

<u>Jules John Magda</u>	, Chair	<u>09/13/2020</u> Date Approved
<u>Swomitra Kumar Mohanty</u>	, Co-Chair	<u>07/30/2020</u> Date Approved
<u>Florian Solzbacher</u>	, Member	<u>09/11/2020</u> Date Approved
<u>Mikhail Skliar</u>	, Member	<u>09/11/2020</u> Date Approved
<u>Anthony Edward Butterfield</u>	, Member	<u>09/13/2020</u> Date Approved

and by Eric G. Eddings, Chair/Dean of

the Department/College/School of Chemical Engineering

and by David B. Kieda, Dean of The Graduate School.

ABSTRACT

Continuous biosensors based upon smart polymer hydrogels have been widely studied because they are relatively biocompatible, inexpensive, and versatile enough to target analytes such as small molecules like glucose or large capsid proteins on virus particles. However, long response times are a typical problem for hydrogel-based sensors. Hence here we investigate a novel transduction method called “resistive channel sensing (RCS)” for hydrogel-based biosensors that is expected to reduce sensor response time. The transduction mechanism uses a potentiostat to monitor the resistance of a microfluidic channel that contains smart hydrogels. Smart hydrogel volume changes in response to changes that occur in the surrounding aqueous solution, such as changes in pH or glucose concentration, thereby modifying the ionic resistance of the microchannel on a microfluidics chip. The development of microfluidic chips includes the design of electrodes and microchannels and is evaluated using solutions with different conductivity to demonstrate proof-of-principle of this sensing approach. The advantages and disadvantages of different electrode materials and geometries with associated channel designs are compared. Smart hydrogel micropillar arrays with various surface area-to-volume ratios integrated into the developed microfluidic chips for RCS show rapid response both to pH and glucose. Further improvement in hydrogel response time can be achieved using an advanced 3D lithography technique known as *two-photon*

polymerization (TPP). The glucose response of smart hydrogel micropillar arrays fabricated using TPP is compared to the glucose response of similar arrays fabricated using traditional darkfield mask lithography.

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CHAPTER 1

INTRODUCTION

1.1 Smart-Hydrogel-Based Microfluidic Sensors

1.1.1 Smart Hydrogels

A smart polymer hydrogel, also called a stimuli-responsive polymer, is a cross-linked polymer network that autonomously and reversibly swells or shrinks in response to some environmental signal, such as change in the concentration of a target analyte [1]. A smart polymer hydrogel responds to changes in the target analyte concentration because it is chemically designed so that it contains molecular recognition elements that reversibly bind to the target analyte. Hence smart hydrogel chemistry is versatile and can be engineered to target many different analytes by adjusting the analyte-binding moieties within the hydrogel. For example, Kataoka and co-workers incorporated boronic acid moieties into smart hydrogels for glucose sensing [2]. However, boronic acid moieties have been known for over a century to strongly yet reversibly bind cis-diols not only on glucose but also on other monosaccharaides such as fructose. Smart hydrogels that contain antibodies and aptamers as pendant groups may provide the potential to enhance selectivity. Miyata et al. used rabbit IgG as an antigen and integrated goat anti-rabbit IgG as molecular recognition elements in the hydrogels [3]. Their hydrogels not only reversibly responded to environmental changes in rabbit IgG concentration but also showed good

selectivity and reversibility. However, antibodies are fragile because they are extremely pH, temperature, and ionic strength sensitive. For this reason, aptamer-based hydrogel sensor development has been a highly researched topic in the past decade [4,5]. DNA aptamers are single strand oligonucleotides that have recently been developed as antibody alternatives, and that have a high selectivity that is comparable to that of monoclonal antibodies, but which are much more stable than antibodies [6]. Bai et al. has shown that this can be achieved by incorporating aptamers into a polyacrylamide-based hydrogel that binds to the protein thrombin at two different amino acid sites [7]. Such aptamer-based hydrogels shrink when they bind to the protein thrombin, because thrombin binding mediates formation of reversible hydrogel crosslinks that cause the hydrogel to shrink in a binding process that is called a “sandwich assay.” The sandwich assay transduction approach can probably be applied to any hydrogel sensor designed to detect large biopolymers such as thrombin. However, such hydrogels whether using chemical moieties, antibodies, or aptamers as molecular recognition elements are usually slow in response, thereby causing swelling response time as a limiting factor for smart hydrogel use in sensing applications. The study of the correlation of hydrogel response time with the dimensions of the stimuli-responsive gel originated with Tanaka and Fillmore in 1979 [8]. Swelling behavior in their study was described to be inversely proportional to the square of the gel radius [1, 8]. Therefore, minimization of hydrogel dimensions will result in faster response of smart hydrogel sensors. For the next generation techniques of hydrogel fabrication using nanotechnology, development of nano/submicron hydrogel synthesis methods are thus of great interest.

1.1.2 Smart Micro-Hydrogel Fabrication Techniques

1.1.2.1 Dark Field Mask Fabrication

2D patterning using dark field masks has been widely used in various applications in fabricating both soft and hard materials [9-11]. Such fabrication often uses UV or visible light to shine through a mask to either solidify or dissolve certain materials in the area that is exposed to UV. Therefore, the wavelength of light imposes a physical limit on resolution. Some studies have shown that 3D structures can be fabricated by stacking up 2D structures layer by layer in multiple steps [12,13]. However, in multistep fabrication alignment is critical and can be time consuming because multiple mask designs have to be alternated among layers of fabrication. Hence, there has been need for technologies that can print true 3D structures in one exposure.

1.1.2.2 Two-Photon Polymerization

Two-photon polymerization (TPP) is a recently developed technique that utilizes a laser for three-dimensional printing. This technology applies two-photon absorption in which both photons have to arrive to the same focal point within a femtosecond interval in order to behave similarly to the traditional single-photon absorption. The theory of two-photon absorption was first introduced in a doctoral dissertation from Maria Göppert-Mayer in 1931 [14]. In contrast to single-photon absorption, two-photon absorption allows photons to have longer wavelengths near infrared, which can minimize Rayleigh scattering and avoid absorption in most of the transparent polymers. This gives a potential to possibly print in 3D with much more complex structures of smart hydrogels in the submicron/nanometer range. Recent research has shown the capability of printing smart hydrogels using TPP. Scaroa et al. successfully fabricated pH responsive hydrogels in

shapes of pyramid and domes using TPP [15]. Kaehr et al. reported that proteins could be photo-crosslinked to a polymer network and the resulting hydrogel was responsive to pH [16]. This study showing that is possible to copolymerize proteins with other pre-gel monomers indicates that we might be able to graft DNA aptamers onto polymer networks in a similar manner.

As described in the previous paragraphs, given that the response time correlates to the reciprocal of radius square of a spherical gels [7], fabrication of smart micro-hydrogels using a traditional dark field mask or through TPP might both result in improvements in response time.

1.1.3 Point-of-Care Sensors

The use of microscopic hydrogels within microfluidic sensors has been investigated in several previous studies [10,11]. Since this type of sensor is small and easy to carry, it is a good candidate for a Point-of-Care (POC) sensor. POC sensors offer the advantages that they are portable, user-friendly, inexpensive, and provide rapid detection of targeted analytes with small sample volumes [17,18]. In addition, POC sensors require no infrastructural equipment and less professional personnel. In conventional disease diagnostics, patients rely on basic healthcare infrastructure [19]. Even though central labs have sensors that provide accurate detection, the cost of each diagnostic test is not only expensive but also time-consuming [17]. This may cause delay of diagnosis due to the prolonged time for testing and transportation. One of the major fields in POC sensors development involves microfluidic techniques. The beauty of microfluidic lies in the small amount of sample volume and the potential of fast response and low cost. Some researchers have even

demonstrated microfluidic sensors with high accuracy and sensitivity in detection of analytes [20,21]. Microfluidic system design includes channel pattern design, materials selection, choice of signal generator, and choice of signal transducer.

1.1.4 Transduction Methods in Smart Hydrogel Sensors

A continuous chemical sensor can be obtained by combining a smart polymer hydrogel with a means for transducing hydrogel swelling changes into electrical signals [22]. Assuming that a smart hydrogel can be designed so that it binds the target analyte, various methods have been studied for transducing hydrogel changes that occur after target analyte binding into electrical signals, including electrochemical, optical, and mass-based methods [23]. Table 1.1 lists different transduction methods that have previously been used with smart hydrogels sensors. In optical transduction methods, light scattering, absorption, or fluorescence changes are used to detect changes that occur when the hydrogel binds the target analyte. Such methods are relatively non-invasive when implanted into the body, but photo-bleaching effects and interference from biomolecules that absorb electromagnetic radiation as similar wavelengths reduce their accuracy. In the mass-based transduction method, analyte binding to the hydrogels induces some change in hydrogel mass or mechanical properties. In order to measure the change in mechanical properties, a relatively large hydrogel is required, hence resulting in long response times [24]. In the electrochemical transduction method, amperometric, potentiometric, or impedimetric changes are used to detect changes that occur as the hydrogels bind with the target analyte. A portable potentiostat that can be operated using a laptop would not only carry out the electrochemical detection but also make the system suitable for point of use.

However, sensors based upon this transduction mechanism are often plagued by noise and interference from molecules with similar redox potentials [25]. Therefore, there has been a strong need to develop a promising microfluidic sensor integrated with an electrochemical transduction method. A microfluidic electrochemical sensing system design includes electrode materials selection, choice of electrode geometry, and choice of carrier solution and working potential range.

1.2 Motivation

A considerable amount of research has already been performed on smart-hydrogel-based sensors, as reviewed in the previous section. However, one can identify a number of technical problems that need to be resolved. The response time and size of hydrogel-based sensors needs to be reduced; hence new transduction methods are needed to achieve this goal.

Microfluidic resistive pulse sensing (MRPS) is a well-studied technique in which the size of nanoparticles can be determined by monitoring the electrical resistance of a microchannel while such particles are passing through [29]. This technique uses the magnitude of sudden resistive pulses when particles pass through an aperture to identify different particles sizes. In a similar fashion, when smart hydrogels in a microchannel change in size in response to environmental stimuli, we might be able to detect a significant resistance change. Thus, we aim to develop a new sensing technique that is similar to MRPS and that can continuously monitor the resistance of a microchannel that contains smart hydrogels. We call this approach resistive channel sensing (RCS).

1.3 Hypothesis

In MRPS, various models have been developed to predict the relative change in resistance that occurs when different sizes of particles pass through a pore [30-32]. With the assumption that particles were electrically insulated and that the current density across the pore cross-sectional area was uniform, they have found that the models can differ from case to case as the ratio of particle diameter to the pore diameter changes [30-32]. However, all the models show that the magnitude of the resistance increases when the volume fraction of the pore occupied by the particle increases. In other words, MRPS can transduce the presence of a nonconductive particle in the pore to an electrical current signal because the pore resistance differs when the particle occupies a specific pore volume. Moreover, the larger the particles, the higher the magnitude increase in resistance.

In RCS, we replace insulated particles that can pass through a pore with smart hydrogels that are stationary and located in a microchannel. We suspect that smart hydrogels are not electrically insulated, however the ionic resistivity of smart hydrogels should be greater than that of the surrounding electrolytic solution because the mobilities of ions are different in the two domains. The resistance of the whole microchannel should thus change when the volume fraction of the gels in the microchannel varies in response to environmental stimuli. For this reason, we hypothesize that RCS will be able to transduce smart hydrogels' volume change to a relative change in the electric current signal because of the resistance change. Given that the conductivity of a solution is the summation of ionic conductivities from each ionic species and that ionic conductivity is proportional to ionic mobility, we also even predict that the conductivity of hydrogels is less than that in the surrounding solution because the ionic mobility should be less in hydrogels than in typical

solutions. As a result, we expect to see the resistance of a microchannel increases when the hydrogels swell in the microchannel.

1.4 Dissertation Outlook

The aim of the research work for this dissertation was to develop a new type of smart hydrogel sensor for POC application which needs to be reliable, rapid in response to environmental stimuli, and require only a small sample size.

Chapter 2 investigates various materials and geometries of electrodes and microchannels in order to motivate the trajectory of the sensor development in this dissertation. We evaluate the reliability of various microchannel designs by reading the ionic current signal observed when introducing different ionic strength solutions into a hydrogel free microchannel along which a small bias voltage is applied.

Chapter 3 presents proof of principle results for a microfluidic pH sensor using RCS with three-point electrodes. This study demonstrates a 2D photolithography technique that can be used for in situ fabrication of pH sensitive hydrogel pillar arrays with high surface area-to-volume ratio in microchannels. A response time improvement of the pH sensor obtained by increasing the surface area-to-volume ratio of the hydrogel is also presented. The chapter was published in the journal *Gels* in 2018.

Chapter 4 is a follow-up manuscript submitted for publication that demonstrates the time-dependent response of microfluidic glucose sensors using RCS with a four-electrode configuration. The use of four electrodes is shown to improve sensor response. The hydrogel pillar arrays have an even higher surface area-to-volume ratio than those studied in Chapter 3. The fabrication process also shows improvements that make it more user

friendly and less expensive.

In Chapter 5, it is shown that 3D structures in a single exposure step can be obtained by applying TPP to a pre-gel solution contain glucose-sensitive monomers, thereby obtaining a microscopic array of glucose-responsive micropillars. A 3D printed glucose sensitive hydrogel in a microchannel that behaves as a flow guide is presented.

Chapter 6 summarizes and discusses the findings and progress presented in this dissertation.

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Table 1.1 Strategies for biomolecule detection using hydrogel-based biosensors

Target	Binding Strategy	Transduction Strategy	Reference
Glucose	Chemical Moiety	Mass-based (pressure sensor)	[24]
Glucose	Chemical Moiety	Electrochemical	[25]
Thrombin	Aptamers	Optical	[26]
Virus	Aptamer	Optical	[27]
Virus	Antibody	Electrochemical	[28]

CHAPTER 2

DESIGN OF MICROFLUIDIC SENSORS THAT USE RESISTIVE CHANNEL SENSING TO TRANSDUCE VOLUME CHANGES OF ARRAYS OF SMART HYDROGEL MICROPILLARS

2.1 Introduction

Resistive channel sensing (RCS) is a new technique similar to a well-studied technique microfluidic resistive pulse sensing (MRPS). MRPS has been used to determine the size of particles by analyzing the magnitude of the spikes of the electrical resistance of a microchannel when the particles pass through [1]. Such sensing technique usually requires an electrochemical cell that contains a microchannel with multiple electrodes to apply a bias potential along the channel, and the channel is filled with an electrolyte solution to create a conductive path [2]. While a particle passes through the sensing region, the particle decreases the cross-sectional area of the microchannel and, thus, generates a resistive pulse that can indicate the size of the particle. In a similar manner, if we replace the particles with stimuli-responsive hydrogels in such an electrochemical cell as mentioned above, the resistance of channel might also differ if the gels reversibly swell or shrink and thereby change the cross-sectional area of the channel. Figure 2.1 demonstrates the concept of RCS.

The sensitivity and reliability of MRPS have encountered some challenges due to underestimation of the particle size when a bias DC potential is applied to the cell. The polarization of ions occurring at the electrolyte-electrode interface, thereby forming electrical double layers (EDL), reduces the potential through EDL so that the real potential across the microchannel is always less than the setpoint at the electrodes [3-4]. However, for RCS, the length of the EDL is typically a small fraction of the microchannel length. Moreover, the reduced bias potential may still be applicable across the microchannel. Hence this phenomenon may not be a problem. However, there are multiple mechanisms that can cause an ionic resistance change for the channel, including the volume change of the hydrogels, transient EDL formation at the electrodes, and faradaic reactions at the electrodes. The transient EDL formation depends on the materials used for the electrodes, the dimensions of the electrodes, the flow rate of solution in the microchannel, and the salt concentration in the microchannel [5]. Faradaic reactions at the electrodes can vary with respect to different electrode materials and ions in the solution. Therefore, a systematic test that can thoroughly characterize or even deconvolute each effect that can potentially overlap with the hydrogel signal is needed.

Here we aim to develop a robust microfluidic hydrogel sensor using RCS. We validate the design of an RCS sensor by checking the pattern of the electric current signal while the microchannel contains solutions with different ionic strengths of phosphate buffered saline (PBS) and is with different bias DC potential applied across. Three different types of materials, gold, silver paste, and composite conductive poly(lactic acid), were used as candidates for the electrodes. Two different patterns of microchannel flow channels were investigated. The durability of the electrodes to corrosion is also included in this chapter.

2.2 Materials and Methods

2.2.1 Fabrication of Substrates with Patterned Electrodes

There were three different types of materials, gold, silver paste, and composite conductive poly-(lactic acid) chosen as candidates for the electrodes. Two configurations, 3-electrode and 4-electrode, were prepared in this study. The details of the electrode configurations for each material is listed in Table 2.1. The silver electrode had silver paste (MG Chemical) stenciled ($1\text{ mm} \times 25\text{ mm} \times 0.04\text{ mm}$) onto the surface of a rectangular piece of polycarbonate ($40\text{ mm} \times 75\text{ mm} \times 0.25\text{ mm}$) from McMaster Carr (McMaster Carr; Elmhurst, USA). The gold electrode used in this study contains two thicknesses, 57 nm and 302 nm, as measured with the profilometer from KLA-Tencor (Model P20; Milpitas, USA). Thin gold-films with titanium used as adhesion layers for both of these gold electrodes were sputter deposited onto standard microscope glass slides obtained from Fisher Scientific (Fisherbrand Superfrost; Hampton, USA). The 302 nm gold electrode was stenciled patterned directly during the sputtering. The 57 nm gold electrode was obtained first sputtering uniformly on glass slides, and then stenciled patterned with a two-step etching process: first gold etchant DI water/KI/I₂ (40:2:1 in mass) to etch gold-films [6] and then 48wt% H₂SO₄ to etch out titanium at 60°C [7]. The composite conductive poly-(lactic acid) (PLA) from Proto-Pasta (Vancouver, USA) was co-printed in 3D with a non-conductive translucent PLA from Sunlu (Perth Amboy, USA) using a commercial two-nozzle-head printer from Flashforge (Model Creator Pro; Jinhua City, China). The electrode designs were created in Solidworks (Version: 2019; Dassault Systèmes SolidWorks Corporation) and then converted to the adaptable format for the 3D printer using the software Flashprint that came with the printer (Version: 3.25.1).

2.2.2 Microchannel Sensing Device Fabrication

The microfluidic sensing devices consist of three layers. First, the bottom layer in the sensing device was a substrate patterned with electrodes as described in the previous paragraph. Second, the middle layer was a patterned polyvinyl chloride (PVC) adhesive film, 50 μm thick, that binds the top layer as well as defines the thickness of the microchannel. Designs of the channel were created in Solidworks (Version: 2019, Dassault Systèmes SolidWorks Corporation) and then cut with a knife plotter (Model CAMEO 2; Silhouette America Inc.). There were two types of channel designed in this study: one had a single rectangular channel that passes across all electrodes, and the other had a side channel in addition to the main channel. The side channel was only in contact with the reference electrode and was separated from the main channel that passes across the rest of the electrodes. Lastly, the top layer was a rectangular piece of polycarbonate with holes punched through to access the microfluidic channels. The fabrication of the 3-layer sensing device and the PDMS tubing connectors were described in the prior work [8-9]. Figure 2.2 shows an illustration of the assembly of a microfluidic sensing device with multi-channels and a 3D printed substrate as an example. Note that the microchannel device that uses silver electrodes as the substrate needs an additional material, polydimethylsiloxane, to seal the 10 μm gap between the electrodes and the top polycarbonate sheet but which is not in contact with the channel region.

2.2.3 RCS of Microfluidic Sensors in Response to Changes in Ionic Strength

All the configurations of electrodes and microchannels are shown in Figure 2.3. We investigate here the ionic current behavior in response to changes in the conductivity in the microchannel by introducing solutions with different ionic strengths. We suspect that the value of the ionic current will be greater when the solution with higher ionic strength is introduced. A syringe pump (Model 780212; KD Scientific Inc., Holliston, USA) was used to withdraw solutions from the reservoirs and passively pump these solutions into the main channel and the side channel of the microfluidic sensors. The main channel was alternately connected between two or three reservoirs that contain PBS with different ionic strengths (150 mM, 100 mM, 50 mM, and 15 mM PBS); the side channel used to control the reference electrode potential contained PBS with a constant ionic strength [10]. The flow rate was varied between 10 $\mu\text{L}/\text{min}$ and 40 $\mu\text{L}/\text{min}$ from experiment to experiment. This flow rate implies a Reynolds number value of less than 200. The ionic current within the main microfluidics channel was measured using a potentiostat from Gamry Instruments Inc. (Model Reference 600; Warminster, USA) with a four-electrode configuration or another potentiostat from PalmSens (Model EmStat3+; Houten, Netherlands) with a three-electrode configuration.

A three-electrode configuration in a potentiostat includes a working electrode (WE), a counter electrode (CE), and a reference electrode (RE), whereas a four-electrode configuration in a potentiostat has an additional working sense electrode (WS). Figure 2.3 (a), 2.3 (b), 2.3 (c), and 2.3 (d) shows the chemical and electrical setup for the case of silver paste electrodes with a single microfluidics channel, conductive PLA electrodes with

multiple microfluidic channels, thin gold-film electrodes with a single channel, and thin gold-film electrodes with multichannels, respectively.

For a three-electrode configuration in a potentiostat, the system operates by applying a small bias voltage between the WE and RE and reading the resulting current across the microchannel between the WE and CE. The Chronoamperometry method was used for the three-configuration electrodes in order to record the ionic current data in PSTrace (Version 5.2, Houten, Netherlands) using a software application that came with EmStat3+. For a four-electrode configuration, the system operates by applying a small bias voltage between the working-sense and the reference electrodes and reading the resulting ionic current across the microchannel between the working and the counter electrode. The Chronoamperometry method was used for the four-configuration electrode system in order to record the ionic current data in the Gamry Instruments Framework (Version 6.25, Warminster, USA), a software application that came with the instrument. A 30 seconds pretreatment with a constant DV voltage of 1 volt was applied before all data collection; various DC voltages were then applied throughout all experiments in this study.

2.3 Results and Discussion

2.3.1 Silver Paste Electrode

Figure 2.4 (a) shows the electrical setup of the microfluidic sensor using silver paste as the electrodes. The microchannel filled with food coloring demonstrates a single channel that passes through all three electrodes. The result in Figure 2.4 (b) was obtained by applying a small bias voltage, 0.3 V, between WE and RE using a potentiostat. The ionic current flowing through the WE and CE and was recorded using the software that came

with the potentiostat. The ionic current responses in 150 mM and 100 mM PBS differ significantly between the first and the second experiments, whereas both of the experiments show similar current values in 15 mM PBS. The inconsistency of the current readouts in the cases of 100 mM and 150 mM PBS might be attributed to the design of three-configuration potentiostat. The potentiostat in three-electrode mode usually does not monitor the potential difference between the WE and CE nor can it be manually confined to a constant value because the surface reaction kinetics of analytes at the working electrode is usually of main interest [11]. In other words, by monitoring the current at the WE and by knowing the potential between RE and WE, the information is sufficient for one who wants to study the standard electrode potential of analytes at the WE. Therefore, using a potentiostat with a three-electrode configuration can monitor the ionic current change that occurs primarily due to reaction at the electrode, and this current is probably only slightly affected by the presence of hydrogel pillars within the microfluidics channel. In addition, the method for silver electrode deposition can only create relatively thick electrodes, ca. 40 μm , and the procedure for the assembly fabrication also needs to be simplified. Thus, we changed the electrode material from silver to gold using sputtering to deposit thin gold films on top of a standard microscope glass slide as the new electrode substrate.

2.3.2 Thin Gold Film Electrodes

Given that the procedure to assemble a device with silver electrodes is so complicated because of electrode thickness, a glass substrate containing sputtered thin gold-film electrodes could substantially simplify the fabrication procedure by getting rid of the channel alignment and the sealing steps. However, the robustness of thin gold film

electrodes and the current response for changes in ionic strengths needs to be investigated. Thin gold-film electrodes with two different configurations were prepared, as indicated in Figure 2.5 (a). One configuration places the WE and RE close together, whereas the other configuration places the CE and RE close together. The microchannels in both devices were alternatively filled with 100 mM PBS and 50 mM PBS at a constant flow rate of 10 $\mu\text{L}/\text{min}$; the applied bias voltage of 0.6 V between the WE and RE was held constant. The ionic current baseline for both ionic strengths was expected to be a horizontal line. However, neither of the configurations gave a stable baseline signal, apparently because both WEs were found to be corroded after two hours of the experiment. Figure 2.4 (b) shows that the configuration that has CE and RE closer together seems to corrode more than the configuration in which the WE and RE are closer. Thus, the corrosion result indicates that the electrochemical system for the former configuration is less stable than for the latter one. In theory, having CE and RE closer should provide us more information concerning the microchannel resistance because, for a three-electrode configuration, the potential is arbitrarily manipulated between WE and RE. According to the design of a three-electrode configuration potentiostat, the system only compensates for the resistance between the CE and RE and assumes zero resistance between the WE and RE [11]. Therefore, having the CE and RE closer together violates this zero-resistance assumption, and hence the system underestimates the compensated resistance of the system. Therefore, although we were seemingly able to obtain a more accurate resistance value for the microchannel, system stability was sacrificed due to electrode corrosion. At this point, although the configuration that has the WE and RE closer can only inform us of the relative resistance of the microchannel, it at least is more stable to corrosion than the other

configuration. As indicated in Figure 2.5 (b), the thin gold films were corroded about halfway through after only two hours of the experiment. Thus, we need more improvement to stabilize the electrodes.

According to the Nernst equation, the potential of an electrode does not only depend on the electrode material but can also vary due to the effective ion concentration in the surrounding solution. Based on our design that contains a single microchannel, the potential of the RE probably changes when solutions with different ionic strength are introduced into the microchannel. As a result, the potentials of both the WE and the CE have to shift in order to satisfy the original bias voltage setpoint. An ideal RE candidate would keep its absolute potential constant so that the deviation potential against other electrodes may be measured [12]. Having a continuous flow passing through the RE can maintain its environmental concentration so that the RE potential will be constant. Figure 2.6 (a) shows an updated version of a new channel design inspired by Wongkaew et al. wherein they isolate the reference electrode within its own channel and continuously pump a solution of constant composition into that channel so that the absolute potential at the RE is always held constant [10]. In our case, we will passively pump the solutions from the reservoirs to both the side and the main channels. The main channel may switch among solutions with different ionic strengths, whereas the side channel uses the same solution to maintain the RE absolute potential. Beside maintaining the potential of the RE electrode, we modified the electrode configuration from three-electrode to four-electrode, which completely separates the electrodes between ones that apply the bias potential and the electrodes that are used to complete the electrical current. Such a configuration that minimizes the current flow between the two electrodes that apply the bias voltage can

effectively diminish the EDL formation so that the effective bias potential applied to the microchannel will be closer to the voltage setpoint [13].

Figure 2.6 (a) shows the electrical setup of the microfluidic sensor using thin gold films as the electrodes. The microchannel filled with food coloring demonstrates that a side channel passes through the RE and that the main channel passes through all three other electrodes. The flow patterns and directions in the channels are also demonstrated in Figure 2.6 (a) by the red and blue dash-line arrows. The result in Figure 2.6 (b) was obtained by applying a small bias voltage, 1 V, between WS and RE using a Gamry potentiostat with a four-electrode configuration, where the side channel contained a continuous flow with 50 mM PBS. The ionic current between the WE and CE was recorded using the software that came with the potentiostat. The fluid flows in both side and main channels were passively driven from the reservoirs and ended up merging to a port close to the CE using a syringe pump with a withdrawing volumetric flow rate of 20 $\mu\text{L}/\text{min}$. The ionic current responses both in the first and the second trials show a good proportionality of the measured ionic current to the solution ionic strength between 50 mM PBS and 100 mM PBS. However, the electrical current observed using 50 mM and 100 mM PBS seems to be ca. 10% off from experiment to experiment. We suspect this deviation was created because the relative flow rates between the side channel to the main channel might slightly differ from experiment to experiment. The current readout for the 150 mM PBS does not quite follow the proportionality of that of 100 mM and 50 mM PBS. Moreover, the current signal of 150 mM PBS does not seem to equilibrate within 15 minutes after switching the sample to 150 mM PBS. The measured ionic current first drifted up slowly and then dropped down rapidly. We suspect the current drift in the 150 mM PBS solution was due to a Faradaic

reaction that occurred [14-15]. Therefore, the results here suggest that using PBS solutions with concentrations less than 100 mM as the conductive solution provide a better chance to see the current change caused by hydrogels volume change.

2.3.3 Conductive PLA Electrodes

Given that our current channel design separates the RE by its own flow channel in order to maintain the constant reference potential, here we aim to move forward to a new way for the electrode fabrication: 3D printing. Previously all the electrodes were deposited on substrates such as polycarbonate sheets and glass slides. In these fabrication methods, the fabricated electrode-substrate surface could not be uniform in thickness. However, 3D printing for electrode fabrication provides a more reliable method for producing electrode-substrate materials of uniform thickness. Figure 2.7 (a) demonstrates the electrical setup of the microfluidic sensor using 3D printed conductive PLA as the electrodes. The microchannel filled with food coloring shows that a side channel passes through RE and a main channel passes through all rest of other electrodes, WE and CE. The flow patterns and directions in the channels are also demonstrated in Figure 2.7 (a) by the red and blue dash-line arrows. The result in Figure 2.7 (b) was obtained by applying a small bias voltage, ± 0.8 V, between WS and RE using Emstat3+ potentiostat with three-electrode configuration, where the side channel contained a continuous flow with 10 mM or 100 mM NaCl. The electric current flowed through WE and CE and was recorded using the software that came with the potentiostat. The operation and the connection setup of the fluid flow was identical to how that was obtained in Figure 2.6 (b). Figure 2.7 (b) shows that the electrical current changes accordingly to the ionic strengths, which was alternating between

50 mM and 100 mM PBS. The absolute value of the current increased while the main channel contained PBS with stronger ionic strength. However, the magnitude of the current change seems to be depending on the absolute potential at the RE. We found that RE with higher NaCl concentration result in smaller current changes than RE with lower NaCl concentration. Moreover, the current readouts that of + 0.8 V and – 0.8 V were not symmetric with respect to zero. These phenomena indicate that, firstly, the applied bias potentials were created relative to the absolute potential at RE and, secondly, the three-electrode configuration could only show the relative resistance change from one ionic solution to the other meaning the proportionality of electrical current to ionic strength could not hold. Some noises and spikes were seen in the current response and we suspect that these might be due to material limitations of the conductive PLA. We found that the resistance of the WE and CE were both to be ca. 200 ohm-cm, which is at least a hundred times greater than silver paste and gold thin-films electrodes. Moreover, leaking of solution occurred quite often because extrusion 3D printing did not provide an ideal sealing between the melted filaments. As the result the yield of fabricated sensors with good quality was less than 50 percent.

2.3.4 Comparison of Ionic Current Transient Response

Using Various Electrode Materials

The development of microfluidic sensors using RCS has moved forward from single channel to a multi-channels design which separates the RE channel from the main channel in order to maintain the absolute potential of the RE. Both three-electrode and four-electrode configurations show ionic current changes in response to changes in

solutions with different ionic strengths and demonstrate the capability of monitoring the resistance of microchannels using RCS. However, the transient current response has thus far not been studied in this chapter. Since the current response involves kinetic steps such as EDL formation, Faradaic reaction, and smart hydrogel swelling, this complicates system response and makes it harder to identify the specific portion of the signal arising from the smart hydrogel volume change. Thus it is important to study the transient current signal in response to ionic strength changes for hydrogel response identification. Figure 2.8 summarizes the results from Figure 2.6 (b) and Figure 2.7 (b) with positive bias voltage applied in the channel. The data in the first ionic strength cycle between 100 mM to 50 mM was taken and normalized using the end point. The time domain was also shifted using the T_{50} time as the middle point. Figure 2.8 shows that all the current responses seem to form an inflection point at T_{50} . Moreover, all the currents reach equilibrium within a 2-minute time span. Therefore, one should be able to recognize the hydrogel response because the hydrogel swelling response fits to first-order or second-order kinetic models that predict no inflection point in the time dependent curve [16-17].

2.4 Conclusions

A microfluidic chip without hydrogels exhibits a more stable response to changes in ionic strength when the chip is designed to have a side channel that connects the reference electrode (RE) to the main flow channel. Microfluidic chip ionic current response was compared between three-electrode and four-electrode systems. It was found that the ionic current readout for the four-electrode configuration was proportional to solution ionic strength, whereas the response of the three-electrode configuration was not. Three types

of electrode materials were investigated. Electrodes made of silver paste provided reasonable values for the ionic current in response to changes in solution conductivity, and also showed no significant degree of corrosion. However, the silver electrodes are necessarily thick, which complicates the fabrication of the sensor assembly. Thin film gold electrodes are easier to fabricate, but they show high degrees of corrosion. This rate of corrosion can be reduced by designing a multichannel microfluidics chip that maintains the potential of the RE at a constant value. 3D printed electrodes using polymers such as PLA containing conductive particles were also investigated. However, the degree of conductivity of this type of electrode was too low. In conclusion, both silver paste and gold thin-films are good candidates for electrodes. Multichannel design that can maintain constant RE potential is necessary for preventing gold thin-film electrodes from being corroded. Both three-electrode and four-electrode configurations are able to monitor the channel resistance. However, the four-electrode configuration gives more consistent results because the four-electrode potentiostat can tightly control the applied bias voltage along the channel.

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Table 2.1 Summary of electrode and microchannel design

Electrode Material	Electrode Configuration	Electrode Deposition	Electrode Thickness	Microchannel Design
Silver paste	3-electrode	Pipetting	40 μm	Single channel
Thin gold-films	3-electrode & 4-electrode	Sputtering	57 nm	Single and multi-channels
Thin gold-films	3-electrode & 4-electrode	Sputtering	302 nm	Single and multi-channels
Conductive PLA	3-electrode	3D printing	0.25 - 4 mm	Single and multi-channels

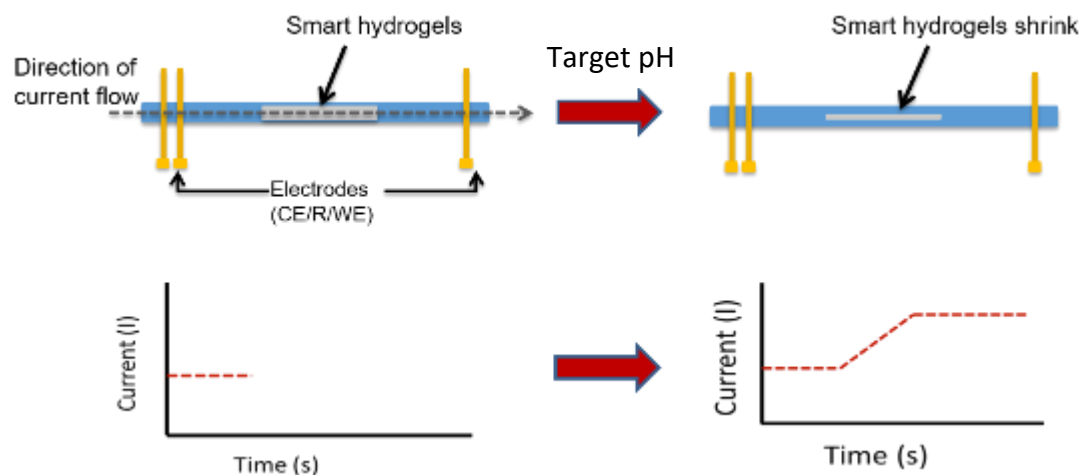


Figure 2.1 Principle demonstration of RCS. Through applying the three-electrode electrochemical setup across the microchannel, the electric current is expected to increase as the target analyte (e.g., pH) approaches to the smart hydrogels and causes it to shrink.

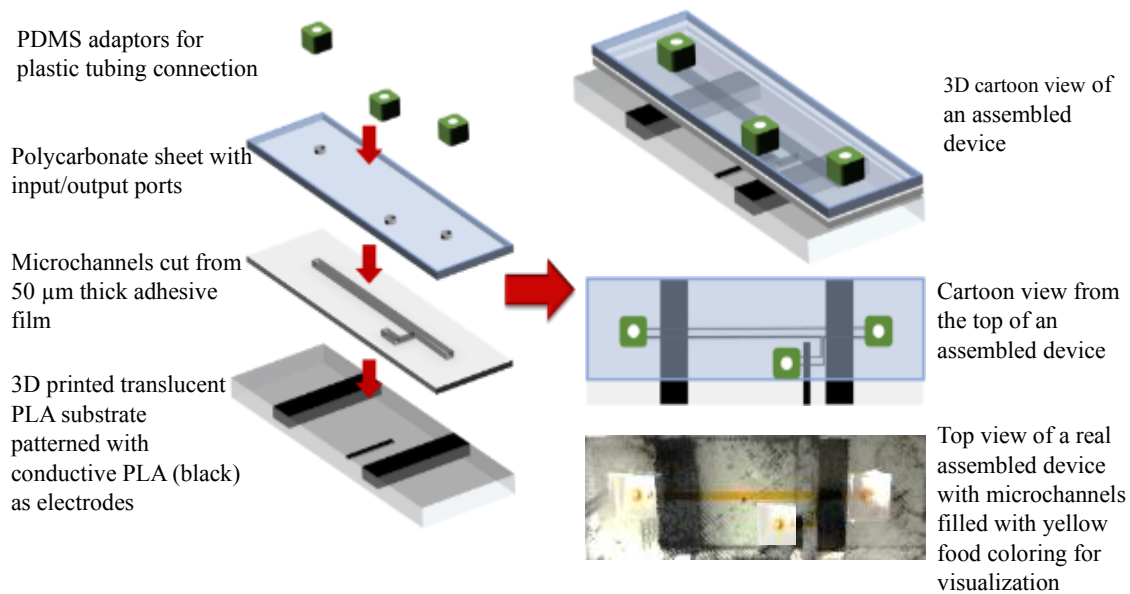


Figure 2.2 Assembly for a microfluidic sensor with multichannel and 3D printed poly-(lactic acid) (PLA) electrode substrate.

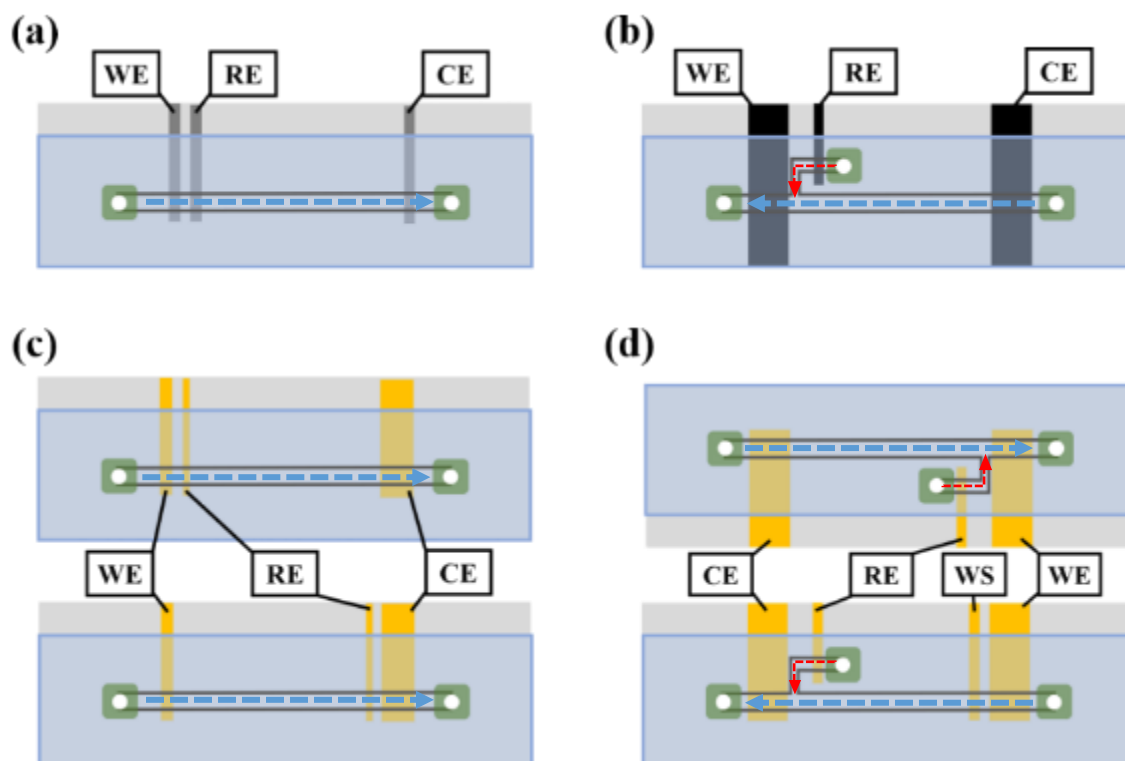


Figure 2.3 The scheme that shows the top-view of the microfluidic sensor and the connections. The flow patterns inside the microchannel are indicated by blue and red dashed lines. The electrode configurations connected to the potentiostat are indicated by black solid lines with labels: working electrode (WE), working sense (WS), reference electrode (RE), counter electrode (CE). (a) Three-electrode configuration with a single channel using silver paste as electrodes. (b) Three-electrode configuration with multi-channels using 3D printed conductive poly-(lactic acid) as electrodes. (c) Three-electrode configuration with a single channel using thin gold-films as electrodes. (d) Three/four-electrode configuration with multi-channels using thin gold-films as electrodes.

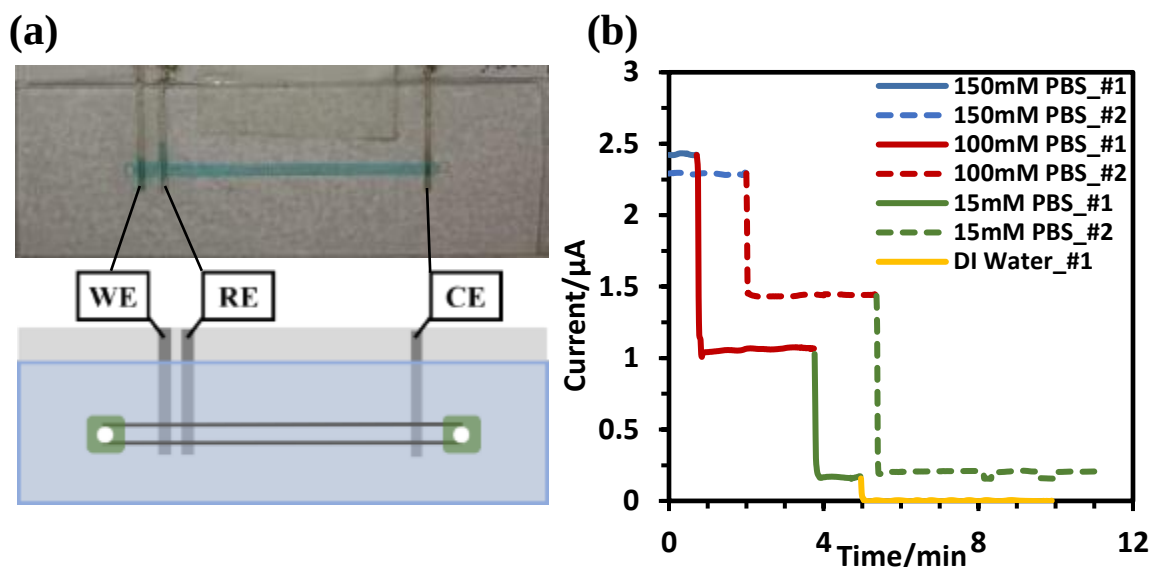


Figure 2.4 The silver paste electrode configurations and the ionic current data with the correlated phosphate buffer saline (PBS) concentrations. The top-view of the scheme and the actual device of microfluidic sensors with silver paste electrodes (Figure (a)). The microfluidic channel of the device is indicated by a diluted blue food coloring for visualization. The electrode configurations connected to the potentiostat are indicated by black solid line with labels: working electrode (WE), reference electrode (RE), counter electrode (CE). Figure (b) shows the ionic current response for reduction in ionic strength in three steps from 150 mM to 100 mM, to 15 mM PBS. The solid curve and the dashed curve represent two replicated experiments, #1 and #2, respectively. Note that the channel in this experiment did not contain hydrogels.

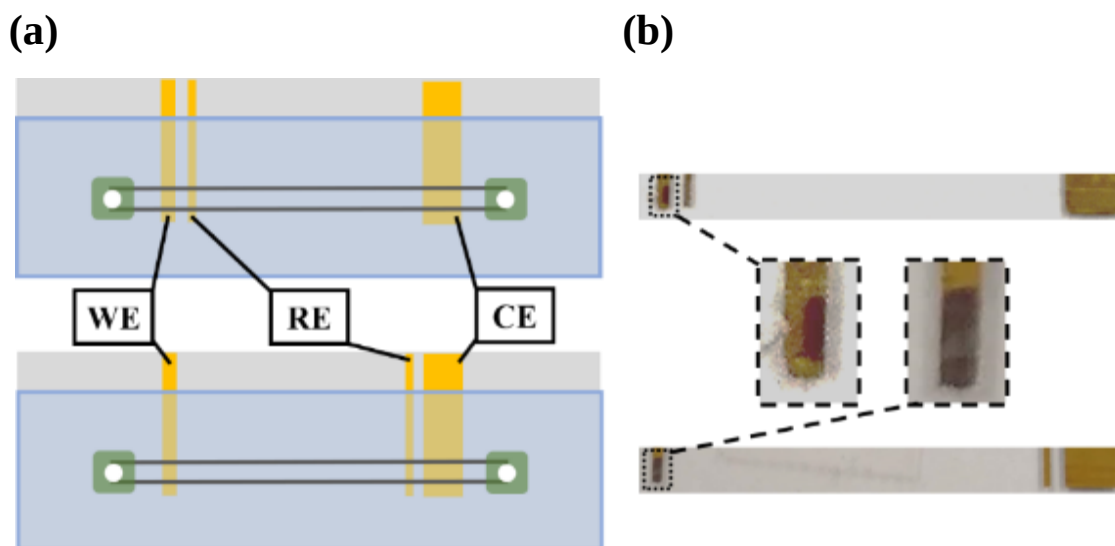


Figure 2.5 The gold electrode configurations and the corruptions of the electrodes. The schemes that show the top-view of the microfluidic sensors in three-electrode configuration and the electrical connections of the thin gold-film electrodes (Figure (a)). The electrode configurations connected to the potentiostat are indicated by black solid lines with labels: working electrode (WE), reference electrode (RE), counter electrode (CE). Figure (b) shows the cropped images of the thin gold-films electrodes, 302 nm, from the two configurations indicated in Figure (a). The WE from both devices was corroded after the electrochemical experiments.

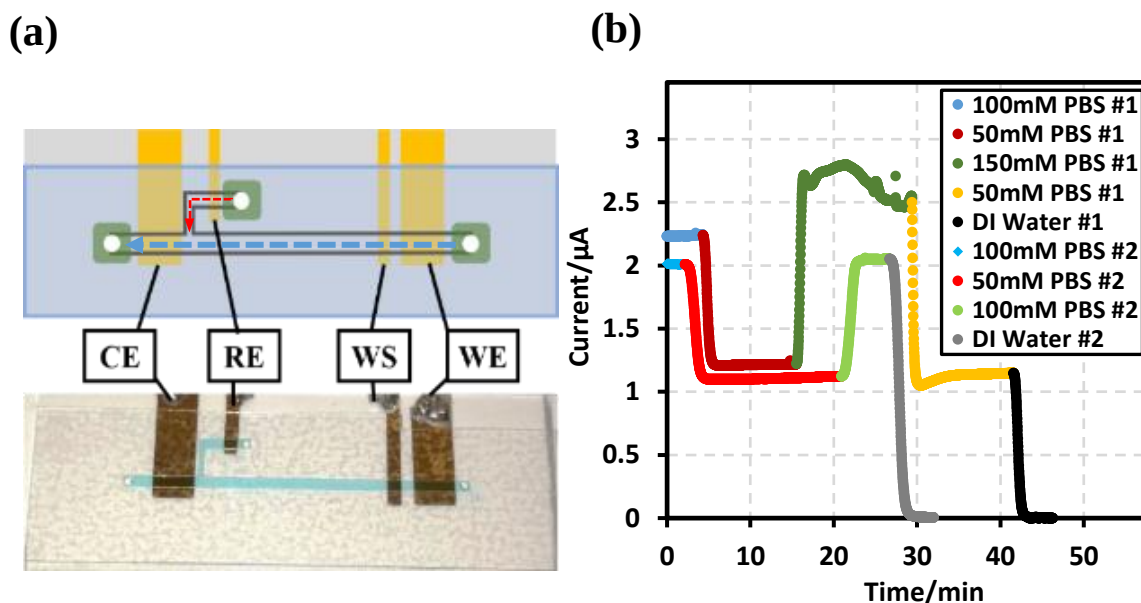


Figure 2.6 The gold electrode configurations and the ionic current data with the correlated phosphate buffer saline (PBS) concentrations. The top-view of a scheme and the actual device of a microfluidic sensor with thin gold-film electrodes with four-electrode configuration (Figure (a)). The red dash-line arrow and the blue dashline arrow indicate the flow patterns in the side channel and the main channel, respectively. The microchannels of the device are indicated by a diluted blue food coloring for visualization. The electrode configurations connected to the potentiostat are indicated by black solid lines with labels: working electrode (WE), working-sense electrode (WS), reference electrode (RE), counter electrode (CE). Figure (b) shows the ionic current response for changes in solution ionic strengths with steps in the following order: 100 mM, 50 mM, 150 mM, 50 mM PBS, then deionized water for the 1st trial. The ionic current response for the 2nd trial with respect to changes in solution ionic strengths begins with 100 mM PBS, followed by 50 mM PBS, 100 mM PBS, then deionized water. The side channel has a continuous flow that contained 50 mM PBS to

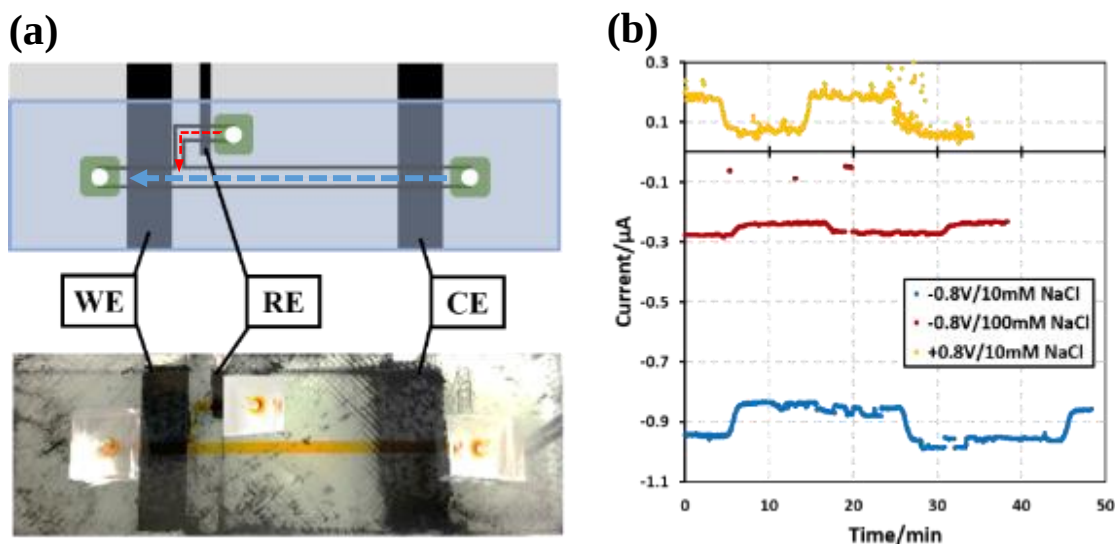


Figure 2.7 The 3D printed electrode configurations and the ionic current data with the correlated NaCl concentrations. The top-view of a scheme and a real device of microfluidic sensors composed of 3D printed electrodes with three-electrode configuration (Figure (a)). The red dash-line arrow and the blue dash-line arrow indicate the flow patterns in the side channel and the main channel, respectively. The microchannels of the actual device are indicated by a diluted yellow food coloring for visualization. The electrode configurations connected to the potentiostat are indicated by black solid lines with labels: working electrode (WE), reference electrode (RE), counter electrode (CE). Figure (b) shows the ionic current response for changes in solution ionic strengths oscillating between 100 mM and 50 mM PBS for two cycles. The side channel has a continuous flow that contained either 10 mM or 100 mM NaCl solutions in order to maintain the absolute potential of the RE and the flow ended up merging into the main channel.

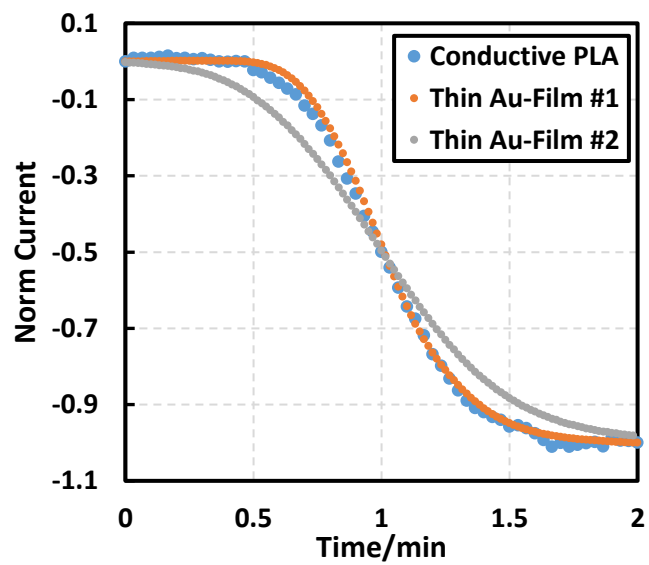


Figure 2.8 Normalized time-dependent ionic current in response to ionic strength that changes from 100 mM to 50 mM PBS.

CHAPTER 3

LOW-COST MICROFLUIDIC SENSORS WITH SMART HYDROGEL

PATTERNED ARRAYS USING ELECTRONIC RESISTIVE

CHANNEL SENSING FOR READOUT

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Article

Low-Cost Microfluidic Sensors with Smart Hydrogel Patterned Arrays Using Electronic Resistive Channel Sensing for Readout

Hsuan-Yu Leu ¹, Navid Farhoudi ², Christopher F. Reiche ², Julia Körner ², Swomitra Mohanty ¹, Florian Solzbacher ^{2,3} and Jules Magda ^{1,*}

¹ Department of Chemical Engineering, University of Utah, Salt Lake City, UT 84112, USA; h.leu@utah.edu (H.-Y.L.); swomitra@chemeng.utah.edu (S.M.)

² Department of Electrical & Computer Engineering, University of Utah, Salt Lake City, UT 84112, USA; navid.farhoudi@utah.edu (N.F.); christopher.reiche@utah.edu (C.F.R.); julia.koerner@utah.edu (J.K.); florian.solzbacher@utah.edu (F.S.)

³ Comprehensive Arrhythmia Research & Management Center, University of Utah, Salt Lake City, UT 84112, USA

* Correspondence: jmagda@utah.edu; Tel.: +1-801-581-7536

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Abstract: There is a strong commercial need for inexpensive point-of-use sensors for monitoring disease biomarkers or environmental contaminants in drinking water. Point-of-use sensors that employ smart polymer hydrogels as recognition elements can be tailored to detect almost any target analyte, but often suffer from long response times. Hence, we describe here a fabrication process that can be used to manufacture low-cost point-of-use hydrogel-based microfluidics sensors with short response times. In this process, mask-templated UV photopolymerization is used to produce arrays of smart hydrogel pillars inside sub-millimeter channels located upon microfluidics devices. When these pillars contact aqueous solutions containing a target analyte, they swell or shrink, thereby changing the resistance of the microfluidic channel to ionic current flow when a small bias voltage is applied to the system. Hence resistance measurements can be used to transduce hydrogel swelling changes into electrical signals. The only instrumentation required is a simple portable potentiostat that can be operated using a smartphone or a laptop, thus making the system suitable for point of use. Rapid hydrogel response rate is achieved by fabricating arrays of smart hydrogels that have large surface area-to-volume ratios.

Keywords: smart hydrogels; microfluidic sensors; UV photopolymerization; fast response time

1. Introduction

A smart polymer hydrogel is a cross-linked polymer network that autonomously and reversibly swells or shrinks in response to some environment signal, such as change in the concentration of a target analyte as for example glucose [1,2]. A continuous chemical sensor can be obtained by combining a smart polymer hydrogel with a means for transducing hydrogel swelling changes into electrical signals [3]. Smart polymer hydrogels can be chemically tailored to selectively respond to many different analytes, but swelling response time is often a limiting factor for their use in sensing applications. Given that the rate of analyte mass transfer is often the rate-determining step in hydrogel response [1], it is likely that shorter response times can be achieved by fabricating smart hydrogels with large surface area-to-volume ratios. Hence, we investigate the potential use of mask-templated UV photopolymerization to produce microscopic smart hydrogel pillars with large surface area-to-volume ratios and, consequently, fast response rates. We introduce a

procedure by which arrays of regularly spaced smart hydrogel pillars can be fabricated inside sub-millimeter channels located within microfluidics devices. For potential use in chemical sensing, microfluidic devices offer advantages such as potentially being low cost and requiring only small sample volumes [4]. We present sensing results obtained using arrays of regularly spaced hydrogel pillars within two different microfluidic channels, with the pillars having surface area-to-volume ratios of 40 mm^{-1} and 13.3 mm^{-1} , respectively. As expected, the sensor response time is shown to decrease with an increase in surface area-to-volume ratio. The use of microscopic pillars within microfluidic sensors has been investigated in several previous studies [5–7]. However, here we propose a novel method for chemical sensing transduction using smart hydrogel pillars that we call resistive channel sensing. In this sensing approach, smart hydrogel pillars are fabricated within the main channel of a microfluidics device. The microchannel is then filled with phosphate buffered saline (PBS) solution to create a conductive path for ionic current. A DC voltage of less than 1.0 volts is applied to the system through the contact pads (the electrodes), which are in contact with the solution, and the induced ionic current through the channel is measured as an electrical current between the electrodes. Once the analyte reaches the smart hydrogel pillars, the pillars shrink or swell, thereby changing the resistance in the main microchannel that results in a change of the measured current. This sensing approach is similar to the well-studied technique known as microfluidic resistive pulse sensing (MRPS), in which changes in the electrical resistance of a microfluidic channel are used to determine the size of nanoparticles that pass through a microfluidic channel [8]. The principal aims of this proof-of-concept study are firstly to determine the feasibility of fabricating microscopic smart hydrogel pillar arrays with large surface area-to-volume ratios inside microfluidics channels, and secondly to determine the reduction of response time that can be attributed to the use of smart hydrogel pillars within microfluidic sensors.

2. Results and Discussion

2.1. Fabrication of Microfluidic Channels

The microfluidic sensing devices were synthesized using a low-cost fabrication approach with the microfluidic channels fabricated employing a computer controlled cutting plotter [9]. The sensing devices consist of 3 layers. The bottom layer in the sensing device was a rectangular piece of polycarbonate ($40 \text{ mm} \times 75 \text{ mm} \times 0.25 \text{ mm}$) with silver paste electrodes (MG Chemical) stenciled ($1 \text{ mm} \times 25 \text{ mm} \times 0.04 \text{ mm}$) onto the surface. The center layer was a polyvinyl chloride (PVC) adhesive film that binds the layers together and that also serves as the microchannel structure. The PVC layer has a thickness of $50 \text{ }\mu\text{m}$. The top layer was another rectangular piece of polycarbonate ($25 \text{ mm} \times 75 \text{ mm} \times 0.25 \text{ mm}$) with holes punched through the top to access the microfluidic channels. The top layer is slightly smaller than the bottom layer to allow access to the electrodes for measurement. To make interfacing with the device simple, PDMS tubing connectors were fabricated in the lab and cored using similar methods in prior work [10]. Figure 1 shows an illustration of the assembly of the microfluidic sensing device. The channel designs were created in AutoCAD (Version: 2016; Autodesk, Inc., San Rafael, CA, USA) and then cut with the knife plotter (Model CAMEO 2; Silhouette America Inc.). The microchannel has a length of 35 mm , a width of 1.6 mm , and a depth of $50 \text{ }\mu\text{m}$.

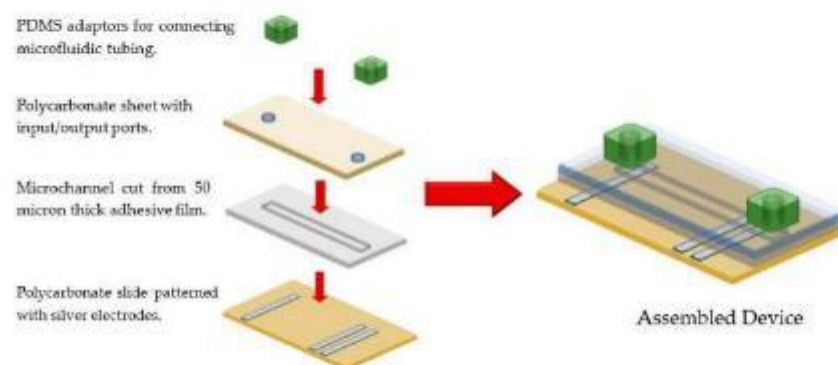


Figure 1. Assembly of microfluidic sensing device for in situ patterning of smart hydrogels.

2.2. UV Photopolymerization of Smart Hydrogel Pillar Arrays within Microfluidic Channels

The pillars were fabricated inside an enclosed microchannel using an in situ photopolymerization technique. Once the 3-layer microfluidic device was cut and assembled, the 13 wt% pre-gel hydrogel solution (see more details in Section 4), which contained 80 mol% acrylamide, 8 mol% 3-acrylamidophenylboronic acid, 10 mol% *N*-[3-(dimethylamino)propyl]methacrylamide, 2 mol% *N,N'*-methylenebisacrylamide and a free-radical photoinitiator was introduced into the microchannel using capillary forces. Subsequently, a dark field photomask with the desired pillar design was placed over the channel. Collimated UV light was used to polymerize the hydrogel to form pillars within the microchannel (Figure 2). After the first photo patterning was complete, the mask was removed and the entire channel containing pre-gel hydrogel solution was flood exposed to the UV light for another quarter of the previous masked exposure time. This step is very important, as it polymerizes a thin hydrogel layer across the channel to enhance adhesion of the hydrogel pillars to the channel and to keep their regular arrangement. When this step was not carried out, it was observed that the patterned pillars did not keep their locations in the channel and were easily flushed out by the surrounding flow. To vary the hydrogel surface area-to-volume ratio, we fabricated two geometrically similar square arrays. The first array has a pillar diameter (as defined by the UV mask) of 100 μm , and a spacing of 200 μm between the centers of the pillars. The pillar diameter was 300 μm in the second array, with a spacing of 600 μm between the centers of the pillars. However, the pillar height (50 μm) and the fraction of the total area occupied by the pillars (19.6%) were the same in both arrays.

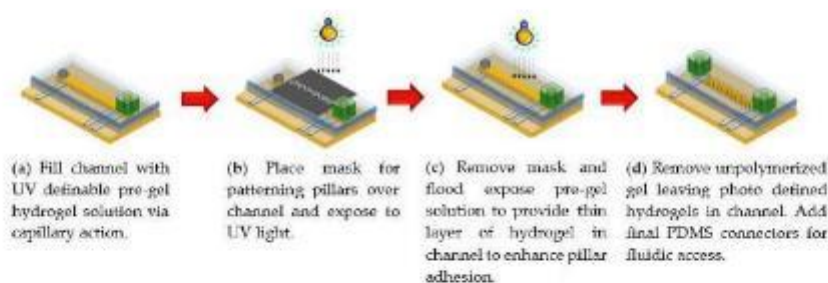


Figure 2. Patterning of hydrogel pillars in microfluidic channel.

2.3. Response of the Hydrogel Pillars to Cyclic Changes in pH

In proof-of-concept response tests, the microfluidics sensor of Figure 2 was subjected to cyclic changes in pH between 7.5 and 10.5. The hydrogel studied here contains both cationic tertiary amines and anionic phenylboronic acid moieties. However, the net hydrogel charge is negative at pH 7.5, and even more so at pH 10.5 [11]. Hence the hydrogel is expected to swell when pH is increased from 7.5 to 10.5. To make this swelling change easier to visualize with an optical microscope, we performed the pH response tests in a low ionic strength saline buffer (1/12× PBS). This reduction in salinity increases the pillar diameter at all pH values, because addition of salt causes hydrogels to shrink by reducing the environmental chemical potential value of water.

Figure 3a shows a micrograph of the array of smart hydrogel pillars as viewed top down in 1/12× PBS buffer at pH 7.5. This micrograph confirms that we succeeded in fabricating a regularly spaced array of smart hydrogel pillars within a microfluidics channel. Figure 3b compares the pillar diameter at pH 7.5 and 10.5. The pillars swell with increase in pH for the reasons discussed above.

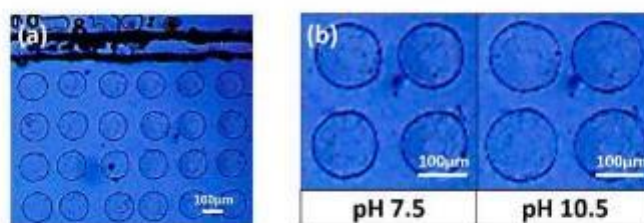


Figure 3. Top-down view of an array of smart hydrogel pillars fabricated by passing UV light through a mask containing an array of circular apertures of diameter 100 μm . (a) Smart hydrogel pillars surrounded by 1/12 $\times$ PBS solution at pH 7.5 (b) Enlarged photograph showing the increase in pillar diameter that occurs when the pH value is increased from 7.5 to 10.5.

When the hydrogel pillar diameter changes due to the change in pH, this changes the value of the ionic current detected by the potentiostat at fixed voltage, as shown by the results presented in Figure 4. Figure 4 shows the time-dependent behavior of the sensor current at fixed voltage as the pH value is periodically changed between 7.5 and 10.5. Results are presented for two different devices, one containing pillars of diameter 100 μm , and the other containing pillars of diameter 300 μm . At the higher pH value, the hydrogel pillars swell, which corresponds to a minimum in the value of the ionic current. The conductance of the microfluidics channel is proportional to both the ion concentration and the cross-sectional area available for current flow. Since the ionic strength was fixed at 25 mOsm/kg in these experiments, the oscillation in current observed in Figure 4 can be attributed to changes in the microfluidics channel cross-sectional area that occur as the pillars shrink and swell. Figure 4 also contains results for the time-dependent *Signal Response %*, defined as

$$\text{Signal Response \%} = \frac{I_{\text{base}} - I}{I_{\text{base}}} \times 100\% \quad (1)$$

where I is the value of the ionic current at a given time t , and I_{base} is the maximum current value measured at times at which the pH value equals 7.5. The value of I_{base} was substantially different for the two devices studied (Figure 4), probably due to variabilities in the device fabrication procedure. Nonetheless, the signal response, calculated as a percentage (Equation (1)) was quite similar for the two devices studied (Figure 4). Most of the pH response of the smart hydrogel studied here occurs near pH 7, because this is close to the pK_a value of PBA inside polyampholytic hydrogels [11]. Hence this sensor could not be used to detect changes in pH between values less than 6, or to detect changes

in pH between values greater than 8 (unless we changed the hydrogel). Based upon the results in Figure 4, we estimate that the sensor has a resolution of about 0.1 pH units near pH 7.

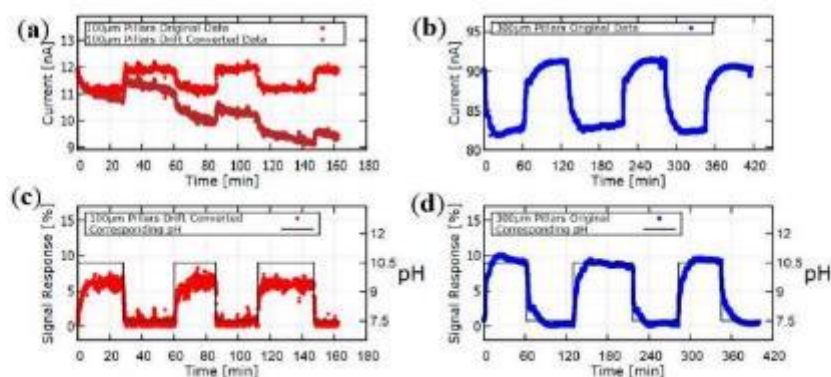


Figure 4. The time-dependent value of the sensor ionic current (a,b) and the Signal Response % (c,d) for periodic changes in pH between 7.5 and 10.5. The pillar diameter as defined by the UV mask was 100 µm (a,c) or 300 µm (b,d). In (c), the Signal Response % has been corrected for baseline drift.

The pH response data in Figure 4 was used to calculate the T_{90} response times of the two sensor devices studied. The results are presented in Figure 5. For both devices, the swelling response time is shorter than the shrinking response time. This may potentially be explained as follows. When a hydrogel starts to shrink, it shrinks first at its outer surface, thereby creating an outer surface film with a low permeability that retards further diffusion of the target analyte into the hydrogel. This, of course, tends to increase the hydrogel response time. For both swelling and shrinking response, Figure 5 shows that the response time is smaller for sensors containing smaller diameter hydrogel pillars. Comparison is made in Figure 5 between T_{90} response times obtained using pillars of diameter 100 µm and diameter 300 µm, with surface area-to-volume ratios of 40 mm^{-1} and 13.3 mm^{-1} , respectively. The increase in the surface area-to-volume ratio by a factor of 3 is observed to reduce the sensor response time, averaged over both swelling and shrinking response times, by a factor of approximately 7.

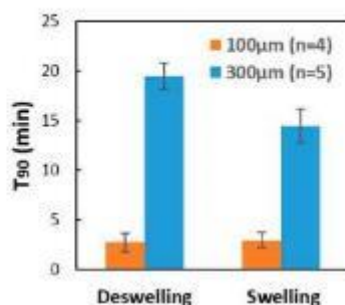


Figure 5. The effect of the pillar diameter upon the T_{90} response time of the microfluidic sensor, as calculated using the response data given in Figure 4. Comparison is made between the results obtained using pillars of diameter 100 µm and 300 µm, with surface area-to-volume ratios of 40 mm^{-1} and 13.3 mm^{-1} , respectively. As expected, the response time is substantially smaller for the sensor that utilizes smaller diameter pillars.

3. Conclusions

In this work, we have demonstrated a method for fabricating low-cost and fast-responding smart hydrogel sensors inside microfluidics channels using soft material microfabrication techniques. The use of photolithographic methods to create micrometer scaled smart hydrogel structures inside a microchannel reduces the cost for this device and removes the need for cleanroom facilities. While in this work we did use a UV source from a mask aligner, a low-cost collimated UV source (such as from Omnicure Inc.) would have been sufficient to create the micropillar arrays.

In the current work, arrays of pH-responsive smart hydrogel pillars were fabricated within a microfluidics channel with surface area-to-volume ratios as large as 40 mm^{-1} . The pH response of these pillars was transduced into an electrical signal using a novel technique termed *resistive diameter sensing*. The electronic signal obtained using this microfluidic pH sensor was shown to be reversible and reproducible. The response time of the microfluidic pH sensor was shown to decrease with increase in the surface area-to-volume ratio of the hydrogel pillars. The fabrication process presented here is a low-cost way to solve a long-standing problem of smart hydrogel analytical devices: namely, their long response times.

4. Materials and Methods

The smart hydrogels studied in this work were both glucose- and pH-responsive. As discussed in reference [12], these hydrogels, which contained 13 wt% of the monomers, were copolymers containing 80 mol% acrylamide from Fisher Scientific (Hampton, NJ, USA), 8 mol% 3-acrylamidophenylboronic acid from Achemo (Hong Kong, China), 10 mol% *N*-[3-(dimethylamino)propyl]methacrylamide from Polysciences Inc. (Warrington, FL, USA), and 2 mol% *N,N'*-methylenebisacrylamide from Sigma-Aldrich (St. Louis, MO, USA). The hydrogels were polymerized via crosslinking copolymerization [1] using lithium phenyl-2,4,6-trimethylbenzoylphosphine from Sigma-Aldrich (St. Louis, MO, USA) as the UV free radical initiator. The light source was a collimated Hg-vapor lamp. While patterning the hydrogel pillars, a dark field chromium photomask with the desired pillars pattern was placed over the channel. Collimated UV light from a mask aligner (Model 206; OAI, San Jose, CA, USA), with an initial intensity of 13.5 W/cm^2 and an exposure time of 5.5 s, was used to polymerize the hydrogel to form pillars within the microchannel. After the photo patterning was complete, the mask was removed. Another 1.5 s of UV exposure was flood applied to the channel itself. This process created a thin film of hydrogel between the pillars to keep the pillars from being flushed away during the introduction of analyte solutions. The UV light intensity decreased slightly from its initial value at the beginning of the experiments. Hence, the exposure time was adjusted accordingly to ensure a constant exposure dose for all experiments. A syringe pump (Model 780212; KD Scientific Inc., Holliston, MA, USA) was used to withdraw analyte solutions from one of two reservoirs (pH 7.5 and pH 10.5 in 1/12× PBS) and into the microfluidic sensors. For the sensor containing the smaller pillars, the syringe pump connection was switched between the reservoirs every 30 min, and the flow rate was $10 \mu\text{L/min}$. For the sensor containing the larger pillars, the syringe pump connection was switched between the reservoirs every 60 min, and the flow rate was $10 \mu\text{L/min}$. This flow rate implies a Reynolds number value of less than 100; the ionic current flow attributable to this flow is of order 1 to 10 nA. The ionic current within the main microfluidics channel was measured using a potentiostat (EmStat3+; using a three-electrode configuration. One electrode pad was connected to the working electrode, while the other two were connected to the counter electrode and reference electrode pads. The system operates by applying a small bias voltage and reading the resulting current across the microchannel. The Chronoamperometry method was used to record the current data in PSTrace (Version 5.2, Houten, The Netherlands), using a software application that came with EmStat3+. A 60 s pretreatment with a constant DC voltage of 0.3 volts was applied before data collection; the same DC voltage was then applied again throughout the entire experimental period.

The targeted solution was introduced into the microfluidic channel using the syringe pump with a flow rate of $10 \mu\text{L/min}$ for at least 20 min before imaging the pillars.

To measure the time-dependent response of the pillar diameter, a digital camera (Model LCM050100KPA; Touptek, Hangzhou, China), installed on a polarizing binocular microscope (Model G508, Unico, Dayton, OH, USA), was used to take photos of the sensor pillar array every 30 s. The syringe pump was used to flow 1/12× PBS into the sensor at a flow rate of 10 µL/min, with the pH value of this solution increasing with time from 7.5 to 10.5 while photos were being taken. The photos were then analyzed using the oval tool from Image J [13] to calculate the diameter of the pillar.

Author Contributions: Conceptualization, S.M., J.M., C.F.R. and R.S.; Methodology, C.F.R., J.K., S.M., H.-Y.L. and N.F.; Validation, C.F.R., H.-Y.L. and N.F.; Formal Analysis, H.-Y.L.; Investigation, H.-Y.L., C.F.R. and N.F.; Resources, R.S., J.M. and S.M.; Writing-Original Draft Preparation, J.M.; Writing-Review & Editing, J.K. and S.M.; Supervision, J.K., C.F.R., J.M., S.M. and R.S.; Project Administration, R.S.; Funding Acquisition, R.S.

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Conflicts of Interest: Florian Solzbacher declares financial interest in Blackrock Microsystems LLC and Sentiomed, Inc.; Jules Magda. declares financial interest in Applied Biosensors LLC.

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CHAPTER 4

RAPID GLUCOSE SENSORS USING SMART HYDROGELS AND ELECTRONIC RESISTIVE CHANNEL SENSING FOR READOUT

4.1 Abstract*

Early stage detection for monitoring disease biomarkers is significant because it has the potential to minimize the spread of diseases and the associated death rate. Early stage detection can be accomplished through the use of inexpensive point-of-use sensors for timely biological and environmental diagnosis without the need for expensive advanced instrumentation. Here we investigate an inexpensive point-of-use microfluidic sensor containing an array of regularly spaced glucose responsive hydrogel pillars with high surface area-to-volume ratio. The hydrogel pillars shrink with increase in glucose concentration, and the degree of shrinkage is transduced into an electrical signal by using a four electrode potentiostat to monitor changes in the ionic current. The response time of the ionic current to changes in environmental glucose concentration is less than five minutes. The ionic current response of the sensor is compared to optical microscope measurements that show the time-dependent swelling or shrinking of the hydrogel pillars in response to changes in glucose concentration. The ionic current responds faster to glucose concentration changes than does the pillar area as measured by optical microscopy,

possibly due to a transient change in ion concentration within the microfluidics channel.¹

4.2 Introduction

A smart hydrogel is a cross-linked hydrophilic polymer network that reversibly swells or shrinks in response to a specific environment signal, for example, a change in glucose concentration [1-3]. A continuous chemical sensor can be fabricated by integrating a smart polymer hydrogel with a means for transducing the hydrogel volume changes into electrical signals [4]. Although smart polymer hydrogels haven shown great potential for use in sensors, their swelling response time usually limits their feasibility in practice. Since the rate of analyte mass transfer into the hydrogel polymer network is typically the rate-determining step [1], one can likely improve hydrogel response times by maximizing their surface area-to-volume ratio.

Large values for the surface area-to-volume ratio of smart hydrogels can be obtained by fabricating regularly-spaced arrays of microscopic pillars within microfluidic channels [5-7]. Our research group has previously published a proof-of-principle paper investigating the behavior of a pH-responsive microfluidics sensor containing regularly-spaced smart hydrogel pillars. The response time of this pH sensor was shown to dramatically decrease with increase in the surface area-to-volume ratio of the hydrogel pillars [8]. Here we investigate a similar microfluidics sensor designed to measure changes in environmental glucose concentration. As in our previous study, we once again use a

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new technique that we call “resistive channel sensing” to transduce changes in hydrogel swelling degree. We use a four-electrode potentiostat to measure ionic current changes in the microfluidic glucose sensor, because we find that it gives more stable results than the three-electrode potentiostat that was used to measure ionic current changes in the microfluidic pH sensor [8].

4.3 Materials and Methods

4.3.1 Fabrication of Gold Thin Film Electrodes

Thin gold films with titanium as adhesion layers were sputter deposited onto standard microscope glass slides obtained from Fisher Scientific (Fisherbrand Superfrost; Hampton, USA). The film thickness was 57 nm measured by the profilometer from KLA-Tencor (Model P20; Milpitas, USA). The resistivity was 8.15 $\mu\text{Ohm}/\text{cm}$ measured by a four-point probe from Magne-Tron Resistivity Test System (Model M800). The Ti-Au films were then stenciled patterned using adhesive vinyl and a computer controlled cutting plotter (Model CAMEO 2; Silhouette America Inc.) [11]. A gold etchant DI water/KI/I₂ (40:2:1 in mass) was used to etch gold films [12] and 48wt% H₂SO₄ was used to etch out titanium at 60°C [13].

4.3.2 Fabrication of Microfluidic Channels

The microfluidic sensing devices consist of 3 layers. The bottom layer in the sensing device is a glass slide patterned with thin gold electrodes as described in the previous section. The two other slightly smaller layers were a patterned polyvinyl chloride (PVC) adhesive film that binds the layers together as well as defines the thickness of the

microchannel, and another rectangular piece of polycarbonate with holes punched through the top to access the microfluidic channels. The fabrication of the 3-layer sensing device and the PDMS tubing connectors was described in prior work [8,14]. Figure 4.1 shows an illustration of the assembly of the microfluidic sensing device. The channel designs were created in Solidworks (Version: 2019; Dassault Systèmes SolidWorks Corporation) and then cut with a knife plotter (Model CAMEO 2; Silhouette America Inc.). The microchannel consists of one main channel and one side channel. The main channel has a length of 50 mm, a width of 1.6 mm, and a depth of 50 μm . The side channel has the dimensions of 7.5 mm x 1.6 mm x 50 μm and is parallel to the main channel, whereas the two channels are connected by a perpendicular channel with the length of 4 mm and the width of 0.5 mm. The channel design here was an updated version to our prior work and was inspired from Wongkaew et al. wherein they isolate the reference electrode with its own channel so that the absolute potential at the reference electrode can always be held constant [8, 15]. This can help stabilize the thin film electrode from being corroded while running experiments.

4.3.3 Synthesis of Glucose Sensitive Hydrogel Pillars

in the Microchannel

The polyampholytic glucose responsive smart hydrogels studied in this work contained phenylboronic acid functional groups that reversibly bind to glucose. As discussed in the study of Lin et al. [16], these copolymer hydrogels contain 80 mol% acrylamide from Fisher Scientific (Hampton, USA), 8 mol% 3-acrylamidophenylboronic acid from Frontier Scientific (Logan, USA), 10 mol% N-[3-

(dimethylamino)propyl]methacrylamide from Polysciences Inc. (Warrington USA), and 2 mol% N,N'-methylenebisacrylamide from Sigma-Aldrich (St. Louis, USA). The hydrogels were polymerized via crosslinking copolymerization [1] using 0.25 wt% of lithium phenyl-2,4,6-trimethylbenzoylphosphinate from Sigma-Aldrich (St. Louis, USA) as the UV free radical initiator. Hydrogel pillar arrays can be fabricated inside an enclosed microchannel in a cleanroom using an in situ photopolymerization technique [14]. However, we here aim to simplify the fabrication process through use of a portable as well as less expensive collimated light source outside the cleanroom. This inexpensive collimated light source contains an UV mercury lamp that emits light through a filter that has a bandwidth of 300-500 nm, a 5 mm tip diameter liquid light guide, and an adjustable spot collimating adapter (Model Omnicure S1000; Excelitas Technologies Corp., Waltham USA). While patterning the hydrogel pillars, a dark field polyester photomask from Fineline Imaging, Inc. (Colorado Springs, USA) with the desired pillars pattern was placed over the main channel. The collimated UV light was then shined through the mask with initial dose of 0.23 J/cm². The UV light intensity was measured using a UV light meter from General Tools & Instruments LLC. (Model UV513 AB; New York, USA) Since the UV mercury bulb light intensity decreased over time, the exposure time was adjusted accordingly to ensure a constant exposure dose for all experiments.

4.3.4 Resistive Channel Sensing for GSH Pillars Response in Glucose

The chemical and electrical setup for the microfluidic glucose sensor are shown in Figure 4.2. The sensing principle is a new technique that we term “chemical resistive sensing.” We assume that the ionic conductivity is lower within the hydrogel pillars than

in the surrounding solution. Therefore, if the hydrogel pillars shrink due to an increase in glucose concentration, this should increase the overall ionic conductivity of the microfluidics channel. A syringe pump (Model 780212; KD Scientific Inc., Holliston, USA) was used to withdraw solutions from the reservoirs and passively pump these solutions into the main channel and the side channel of the microfluidic sensor. The main channel was alternately connected to two reservoirs with differing glucose concentrations (1mM and 10mM glucose in 0.1M phosphate buffered saline, PBS); the side channel used to control the reference electrode potential contained 0.1M PBS without glucose [15]. All the solutions were adjusted to pH 7.4 with an ionic conductivity of 9.3 mS/cm at 20°C. The glucose sample stream was switched between the two reservoirs every 20-30 minutes, and the flow rate was 20 μ L/min. This flow rate implies a Reynolds number value of less than 100. The ionic current within the main microfluidics channel was measured using a potentiostat from Gamry Instruments Inc. (Model Reference 600; Warminster, USA) using a four-electrode configuration. Three of the four electrodes were in contact with the main channel whereas the fourth one, the reference electrode (RE), was only in contact with the side channel. In the main channel, one electrode pad was connected to the counter electrode (CE) at one end, while the other two were connected to the working electrode (WE) and the working-sense electrode (WS) pads at the other end. The system operates by applying a small bias voltage between the working-sense and the reference electrodes and reading the resulting current along the microchannel between the working and the counter electrode. The Chronoamperometry method was used to record the current data in the Gamry Instruments Framework (Version 6.25, Warminster, USA), a software application that came with the instrument. A 30 seconds pretreatment with a constant DV voltage of 1

volt was applied before data collection; the same DC voltage was then applied again throughout the entire experimental period.

Changes in the diameter of the pillars in response to changes in environmental glucose concentration were measured with a polarizing binocular microscope (Model G508, Unico, Dayton, USA) using a digital camera (Model LCMOS05100KPA; ToupTek, Hangzhou, China). Photos were simultaneously captured every 30 seconds while the potentiostat was simultaneously monitoring the ionic resistance of the channel. The objective lens was a PLAN 20/0.4 POL 160/0.17 (magnification 20 $\times$). The photos were then analyzed using the five-point elliptical tool from the software ToupView (Version1; ToupTek, Hangzhou, China) that came with the digital camera in order to calculate the area of the pillars. The software was calibrated using a standard microscope stage calibration slide.

4.4 Results

4.4.1 Microscopic Response of the Hydrogel Pillars to a Cyclic Change in Glucose Concentration

The microfluidics sensor was subjected to cyclic changes between two different glucose concentrations, 1 mM and 10 mM, in 0.1 M PBS. Figure 4.3(a) and 4.3(b) compares microscope images of the pillar array of smart hydrogels as viewed top down in 1 mM glucose and 10 mM glucose in 0.1 M PBS buffer, respectively. These images confirm that we have succeeded in fabricating a regularly spaced array of smart hydrogel pillars of diameter 50 microns within a microfluidics channel using commercial polyester masks and an inexpensive collimated light source. The pillar array is 20 mm long and the

fraction of the total microfluidics channel area occupied by the pillars is 12.6%. The pillars show decrease in diameter as the glucose concentration increases. This is consistent with the swelling response for smart hydrogels of this composition reported by Lin et al. [16]. Figure 4.3(c) shows the time-dependent pillar area shrinking ratio result in response to changes in glucose concentration. An asymmetric response time behavior is observed with the T_{90} response time of shrinking 11 minutes whereas the T_{90} of swelling is ca. 25 minutes.

4.4.2 Ionic Current of the Microfluidic Device in Response to Changes in Glucose Concentration

When the volume of the hydrogel pillars changes due to changes in glucose concentration, this also changes the readout of the ionic current detected by the potentiostat at a fixed potential 1 V, as shown by the results presented in Figure 4.4 (a) and 4.4 (b). Figure 4.4 (a) shows the time-dependent behavior of the ionic current at fixed potential as the glucose concentration is alternated between 1 mM and 10 mM. The result represents the device that contains 50 μm pillars in diameter with a distance of 50 μm between the surfaces of the pillars. Figure 4.4 (b) shows the correlation between the drift converted data and the corresponding glucose concentrations. The electrical current reversibly echoes once the glucose concentration changes from one to the other. A distinct current spike was captured right after the glucose concentration changed from 10 mM to 1 mM in the second cycle. Minor perturbations of electrical current signal consistently re-occur at ca. 4 minutes before the current has any significant change and overlap with the glucose concentration turning points. Figure 4.4 (c) and Figure 4.4 (d) compares the result between microscope and electrical current, which the image capturing and the current monitoring were operated

simultaneously. The ionic current is observed to exhibit a faster response time both in increasing and decreasing glucose concentration.

The glucose response data in Figures 4.4(c) and 4.4(d) was then used to calculate the T_{90} response time. Figure 4.5 compares the observed sensor response times obtained from ionic current and microscopy measurements. Ionic current response rates are clearly much faster: the T_{90} response time of shrinking from the ionic current signal is only 4 minutes, whereas the T_{90} of swelling is only 6 minutes.

4.5 Discussion

The polyampholytic hydrogel studied here that contains the anionic phenylboronic acid moiety and the cationic tertiary amine moiety has been shown to shrink in volume as the glucose concentration increases in PBS at physiological pH and osmolarity [16-19]. Optical images of the pillars in Figure 4.3 (a) and Figure 4.3 (b) show decrease in diameter as the glucose concentration increases as expected. The asymmetric response time behavior between the shrinking and swelling of the hydrogel shown in Figure 4.3 (c) is in agreement with previous publications [16, 21, 22]. However, the response times reported here are much shorter. We suspect it is due to a significant increase of the surface area-to-volume ratio of the hydrogel pillars, which was equal to 80 mm^{-1} in the current study. With respect to the ionic current response, we found that the signal of the glucose sensor closely correlates to change in glucose concentration. As described in our previous work, the current measured in the potentiostatic mode represents the conductance of the microchannel, which is proportional to both the conductivity of the buffer and the cross-sectional area available for ionic transport [8]. Here we fixed the conductivity of all the

reservoirs at 9.3 mS/cm at 20°C. Hence the current response in cycles observed in Figure 4.4 (a) can be attributed to the changes of the cross-sectional area of the microchannel due to the shrinking or swelling of the hydrogel pillars. As expected, when the glucose concentration decreases, the current decreases due to hydrogel swelling. This trend of cyclic electrical current pattern has the same agreement with the microscopic result as indicated in Figure 4.4 (c) and Figure 4.4 (d). However, the response time of the ionic current is much faster than the microscope results by factors of ca. 4 in swelling and ca. 3 in deswelling. The Donnan equilibrium model may be used to postulate a possible reason why the ionic current response of the sensor is faster than the swelling response observed in micrographs [20,23]. When the glucose concentration increases, the net positive charge on the polyampholytic hydrogel decreases, and the hydrogel shrinks. Hence the hydrogel releases free counterions into the surrounding solution, which tends to increase the ionic conductivity of the microfluidic device by an amount in addition to that which can be attributed to hydrogel shrinkage. As the result, both the ion concentration and the cross-sectional area available for current flow transiently increase, and the two effects reinforce each other thereby increasing accelerating the response rate of the sensor. A distinct current spike in Figure 4.4 (b) was detected right after the glucose concentration changed from 10 mM to 1 mM in the second cycle. We suspect that the spike comes from the release of extra ions from the dead volume at the junction of plastic tube and PDMS adapter. The diverging flow from plastic tube to PDMS can occasionally form air bubbles and dead volumes. Minor perturbations in Figure 4.4 (b) occur repeatedly ca. 4 minutes before each of the glucose concentration turning points. The perturbations most likely arise from the operation while switching between the reservoirs; the time delay of the response signal can

be attributed to the travel time of the solution from the reservoir to the sensor.

4.6 Conclusions

In this work, we have demonstrated an improved method for fabricating low cost and fast responding microfluidic glucose sensors. We successfully created glucose responsive hydrogel pillar arrays that have large surface-area to volume ratio of 80 mm^{-1} without using cleanroom facilities, thereby obtaining glucose sensors with response times of order 5 minutes. The readout from these sensors is an ionic current obtained from potentiostat measurements, which includes contributions both from the volume change of the pillars and from the ionic exchange between the pillars and the surrounding solution. Time-dependent values of the ionic current were also compared to optical measurements of changes in the pillar diameter.

4.7 References

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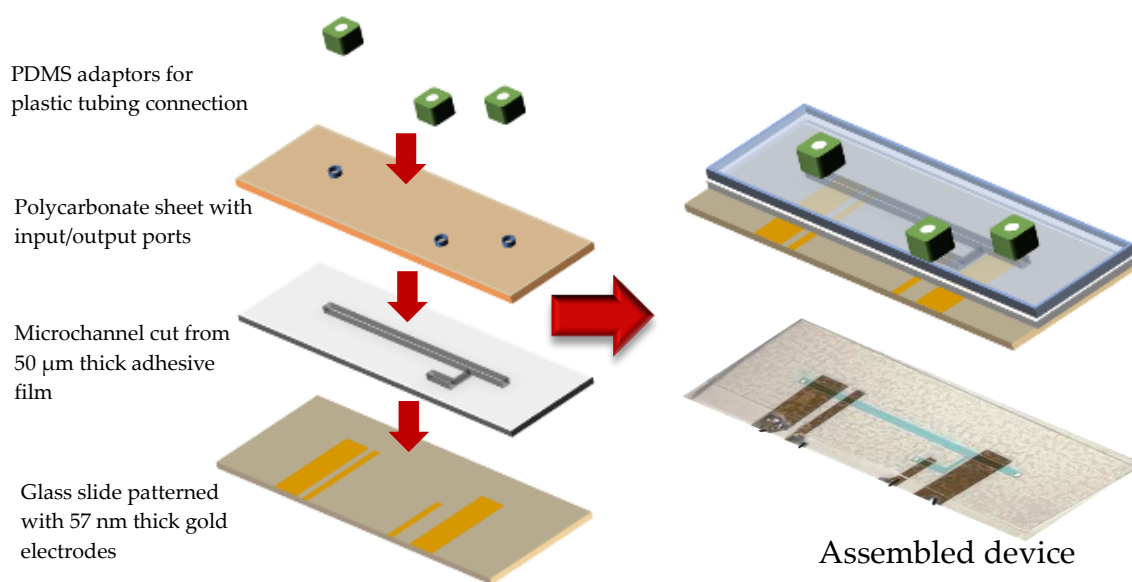


Figure 4.1 Assembly for microfluidic sensor without smart hydrogel pillar arrays. The pillar arrays patterning using photolithography was described in detail in the prior study [8].

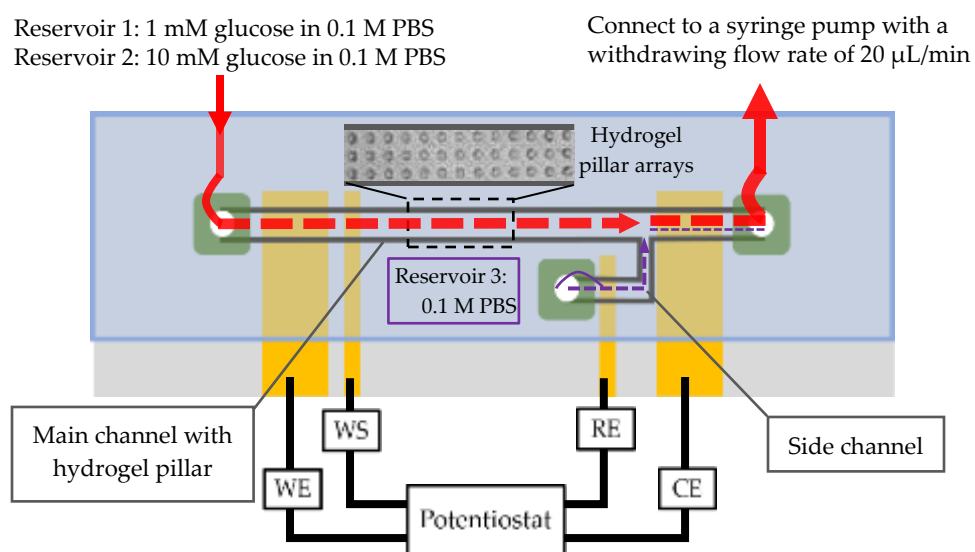


Figure 4.2 The scheme that shows the top-view of the microfluidic sensor and the connections. The flow patterns inside the microchannel are indicated by red and purple dashed lines. The connections between the reservoirs and the microfluidic device are indicated by red and purple solid lines. The electrode configurations connected to the potentiostat are indicated by black solid line with labels: working electrode (WE), working sense (WS), reference electrode (RE), counter electrode (CE).

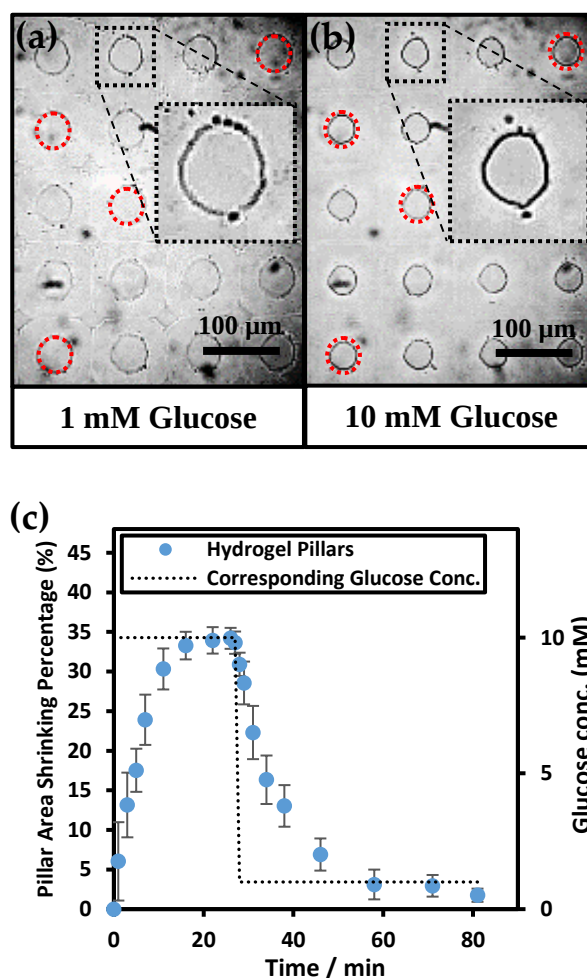


Figure 4.3 Top-down view of an array of smart hydrogel pillars and the time-dependent value of the pillar area percentage change with respect to change in glucose concentration. The smart hydrogel pillars were fabricated by passing UV light through a mask containing an array of circular apertures of diameter 50 μm . (a) Smart hydrogel pillars surrounded by 1 mM glucose in 0.1 M PBS solution at pH 7.4 at equilibrium. (b) Smart hydrogel pillars surrounded by 10 mM glucose in 0.1 M PBS solution at pH 7.4 at equilibrium. (c) The time-dependent pillar area shrinking percentage as obtained by averaging the values among four randomly selected pillars as indicated in Figures (a) and (b) at different times. All the values were then normalized to the value at $t = 0$, which was measured in 1 mM glucose. Error bars denote the standard deviation for four selected pillars indicated in Figures (a) and (b).

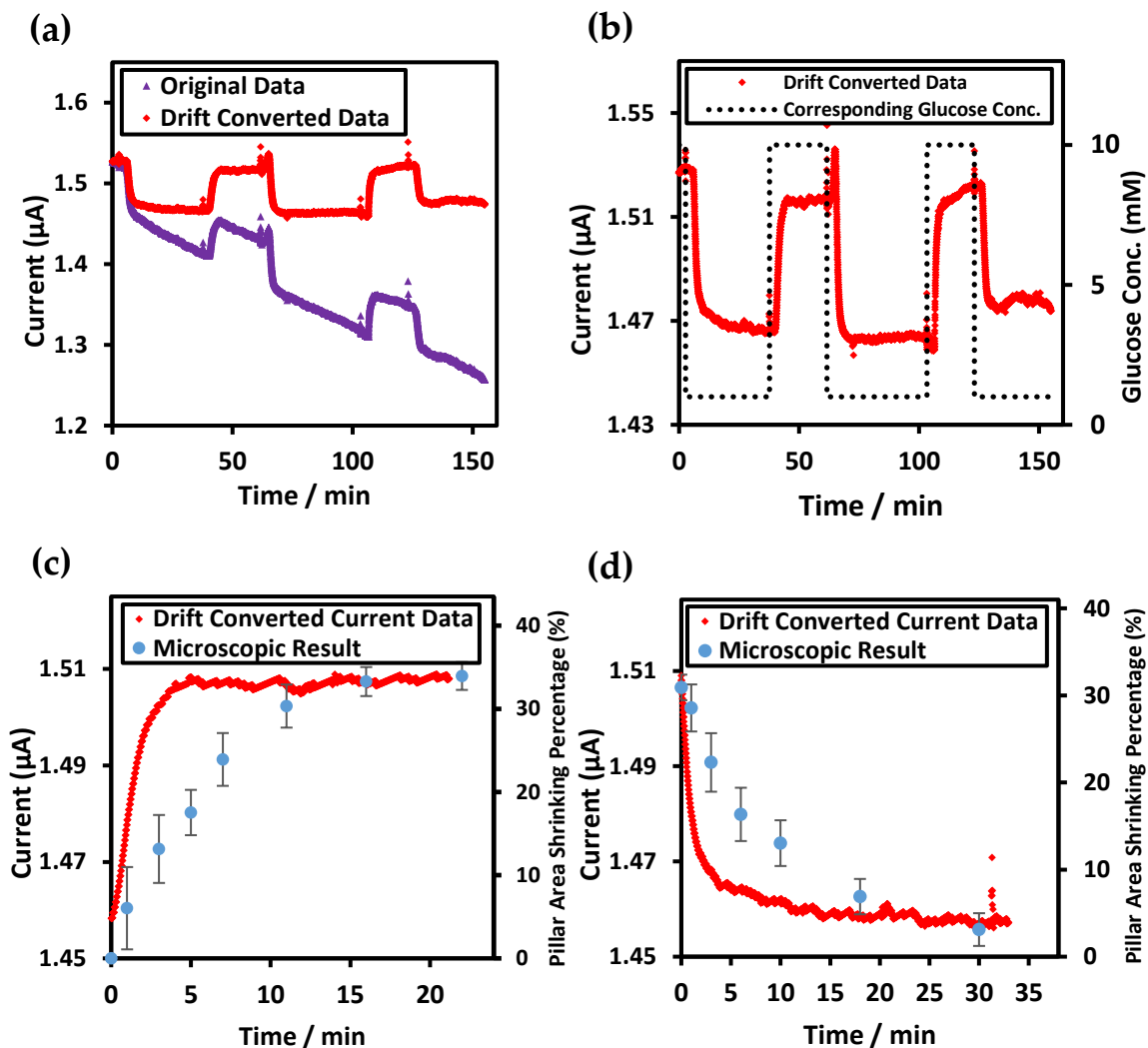


Figure 4.4 The time-dependent value of the sensor ionic current. Figures (a) and (b) show for cyclic changes in glucose concentration between 1 mM and 10 mM. Figures (c) and (d) compare the equilibrium responses to changes in glucose concentration for the ionic current and the pillar area measured from microscope images. Figures (c) and (d) represent the case in increasing and decreasing glucose concentration, respectively. The ionic current is observed to exhibit a faster response time than the optical response in both cases.

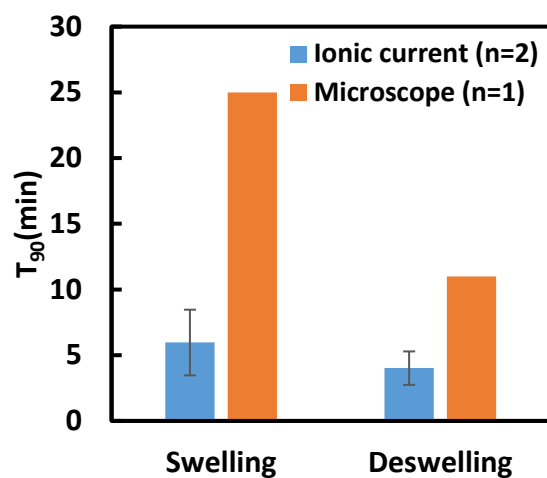


Figure 4.5 Comparison of the T_{90} response time of the microfluidic glucose sensor as calculated using optical and potentiostat measurements using the data given in Figures 4.3 and 4.4. The response time is substantially shorter as measured with the potentiostat.

CHAPTER 5

GLUCOSE SENSITIVE HYDROGEL SYNTHESIS USING TWO-PHOTON POLYMERIZATION

5.1 Introduction

A smart, or stimuli-responsive hydrogel is a three dimensional (3D) crosslinked polymer network that can uptake large amounts of water as well as autonomously swell or shrink in response to specific environment signals [1-3]. Smart hydrogels are able to respond to many analytes [2]. However, the swelling response times of smart hydrogels are usually large, which possibly can be due to the rate of analyte mass transfer into the polymer network [1]. Therefore, there has been a strong need for techniques that can significantly downscale hydrogel dimensions and decrease the critical mass transfer length for response time improvement.

Two-photon polymerization (TPP) is a direct laser writing technique that allows 3D lithography with resolutions on the order of 100 nm [4]. Femtosecond laser pulses trigger a photopolymerization reaction at a tightly focused location. The quadratic dependency of two-photon absorption confines the polymerization at an ellipsoidal focal point (termed as a voxel) [5]. The nonlinear absorption near infra-red allows the laser to focus deep into a material without unwanted scattering and absorption. Thus, TPP enables more complex hydrogel structures to be printed in 3D than traditional dark field mask 2D patterning.

Nanoscribe Photonic Professionals GT2 is the only commercially available tool that utilizes TPP for 3D printing of nanostructures. Nanoscribe takes in CAD models, and prints with the help of a piezo controlled stage and Galvo controlled beam [6]. The printing does not require any support material.

There are a few factors that can challenge the printability and space resolution of hydrogels during TPP, including evaporation of solvent and local thermal and mass driven flow [7]. One can avoid these problems by using less volatile solvents and increasing the monomer concentration and the viscosity of the pre-gel solution. Some studies show the success in fabricating 3D stimuli-responsive hydrogel using TPP. However, to our best knowledge, TPP-printed hydrogels have been fabricated that respond to pH and ionic strength but not to biological molecules or biomarkers [7-9]. In this study, we aim to investigate the key parameters, monomer concentration and viscosity of pre-gel solutions used to fabricate glucose-sensitive hydrogels using the printability of TPP using the Nanoscribe. Moreover, the swelling ratio of TPP hydrogels in response to glucose is studied as evaluations for print quality. A functional valve in a simple 2D microfluidic channel using a TPP hydrogel was demonstrated to show the potential of writing nano/sub-micro 3D smart hydrogels in-situ in a more complex system.

5.2 Materials and Methods

5.2.1 Materials

The monomers used for preparation of the gels were obtained as follows: acrylamide (AAM, Fisher Scientific), N,N-methylenebisacrylamide (BIS, Sigma-Aldrich), 3-acrylamidophenylboronic acid (3-APB, Frontier Scientific, Logan, UT), and N-(3-

dimethylaminopropyl acrylamide (DMA, Polysciences). The monomers were used as received. Lithium phenyl-2,4,6-trimethylbenzoylphosphinate (LAP, Sigma-Aldrich), poly(ethylene glycol) (PEG, Mn = 8000, Sigma-Aldrich), KOH (Fisher Scientific), D(+)-glucose (Sigma-Aldrich), dimethyl sulfoxide (DMSO, Fisher Scientific), 4-(2-Hydroxyethyl)piperazine-1-ethanesulfonic acid (HEPES, Sigma-Aldrich) were also used as received. The apparatus for TPP is Nanoscribe photonic professionals GT systems (Nanoscribe, Boston, USA).

5.2.2 Preparation of Glucose Sensitive Hydrogel Pre-Gel Solutions

A glucose sensitive hydrogel (GSH) containing AAM/3-APB/DMA/BIS with mole ratio of 80/8/10/2 was synthesized by free radical crosslinking copolymerization. This composition was the same as in Tierney et al. [10], with the exception that we used a different kind of photo initiator, LAP, for polymerization and a much higher monomer concentration. In brief, a stock solution that contained 67.8 wt% of AAM and BIS with a mole ratio of 40:1 was dissolved in PEG-thickened HEPES buffer. The buffer contained 50 wt% of PEG and 1mM of HEPES in deionized water and was adjusted to pH 8 using 1M KOH. In order to dissolve 3-APB into the pre-gel solution, ca. 5-20 vol% of DMSO was added into the solution depending on the designated monomer concentration. Appropriate amounts of the stock solution and DMA were then added and mixed in the vial. In some cases, additional quantities of PEG were added to the vial to enhance the viscosity of pre-gel solutions. Lastly, LAP was introduced to 1 wt% of the pre-gel solution. Table 5.1 summarizes the compositions of GSH in this study.

5.2.3 Viscosity Measurements of Pre-Gel Solutions Using Rheometer

A controlled-stress rheometer (DHR-2, TA Instruments) with cone-plate fixtures was used to measure the viscosity of pre-gel solutions. The sample temperature was controlled at room temperature 25°C using the lower Peltier plate on the rheometer. We used the standard smooth cone-plate fixtures (40 mm diameter, 2.009° cone angle, 48 μm truncation gap; standard Peltier plate) to oscillate the pre-gel solution in shear with constant shear stress amplitude of 0.5 Pa from 0.1 rad/s to 100 rad/s. Five data points were collected per decade of frequency. The reported viscosity in this study was the average among five data points in the first decade of frequency.

5.2.4 TPP for GSH Synthesis Using the Nanoscribe

Hydrogels were fabricated using the TPP technique with Nanoscribe photonic professional GT systems. The system contains an IR laser, which can generate femtosecond laser pulses with a center wavelength of 780 nm. A 63 x 1.4 NA objective lens is used in Dip-in Laser Lithography (reference) mode. The objective is immersed in immersol 518F oil, which has a matched refractive index with borosilicate glass. The laser beam is focused on the pre-gel solution through the 170 μm thick glass substrate. The advantage of this mode is that the photo resin is separated by the cover glass substrate, which protects the expensive objective lens from any chemical reaction induced damage. However, the lens's working distance is 340 μm , which is further limited by the thickness of the substrate. Hence the maximum structure height achievable with this mode is 170 μm .

Seventy percent of the calibrated 50 mW laser power with scaling factor 1.0 was

used for the hydrogel fabrication process. The effective laser power is calculated by multiplying the percentage of laser power with the scaling factor. The exposure dose is roughly inversely proportional to the square root of the scan speed. The exposure dose can be increased either by increasing the laser power or by reducing the scan speed. A scan speed of 90 $\mu\text{m/s}$ was chosen for this process.

The intended structure was drawn in CAD and later converted to General Writing Language code using DeScribe 2.5.4 software to make readable for the Nanoscribe system software (NanoWrite 1.8.3). The structure is generated layer by layer with the help of a Galvanometric mirror scanner for x-y scan and a high-resolution piezo-controlled stage movement in the z-direction.

5.2.5 Integration of TPP-Printed GSH into the Microfluidic Channel

A glass cover slip that contained TPP-printed GSH as described in the previous section was then integrated into a microfluidic channel assembly for further gel-conditioning steps and responsiveness tests. The assembly consists of 3 layers, whereas the bottom layer in the sensing device was a piece of glass cover slip with TPP-printed GSH; the rest of the two layers as well as the PDMS tubing connectors fabrication steps were described in the prior work [11, 12]. The TPP printed GSH was located in the middle of micro-channel, which has a length of 19 mm, a width of 3 mm, and a depth of 50 μm .

The microchannel for flow guide demonstration contained two 1 mm wide U-shaped channels upside down to one and the other, where a 1 mm x 0.5 mm rectangle was placed in between as the junction. The GSH valve was then TPP in-situ synthesized at the junction.

5.2.6 Evaluation of GSH Response in Glucose

Using the Optical Microscope

A syringe pump (Model 780212; KD Scientific Inc., Holliston, USA) was used to introduce solutions from the reservoirs and passively pump solutions into the channel. The TPP-printed GSH samples were subjected to three cycles of ionic strength change between 0.16 M and 0.05 M phosphate buffered saline (PBS) for further cleaning, and were then stored in 0.16 M PBS until testing.

During the glucose test, the channel was alternately connected to two reservoirs with different glucose concentrations (0mM and 10mM glucose in 0.05M PBS). All the solutions were adjusted to pH 7.5 with an ionic conductivity of 4.6 mS/cm at 20°C. The glucose sample stream was switched between the two reservoirs every 40-50 minutes, and the flow rate was 20 μ L/min.

Changes in the diameter of the pillars in response to changes in environmental glucose were measured with a polarizing binocular microscope (Model G508, Unico, Dayton, USA) using a digital camera (Model LCMOS05100KPA; ToupTek, Hangzhou, China). Photos were simultaneously captured every 30 seconds. The objective lens was a PLAN 20/0.4 POL 160/0.17 (magnification 20 $\times$). The photos were then analyzed using Image J [13].

5.3 Results and Discussion

5.3.1 GSH Pillars Swelling Ratio in Response to Glucose

All 10 pre-gel solutions were brought to Nanoscribe for TPP fabrication. However, we found that GSH₇₅₋₁₅ was not printable because recrystallization occurred in the pre-gel

solution during the printing. We believe that the recrystallization was due to reaching the monomer solubility limit. GSH₆₀₋₉ using the laser power and the scan speed given in Section 5.2.4 resulted in no polymerization. Stronger laser power and slower scan speed were tried to provide more exposure dose, but the pre-gel solution then formed bubbles, which resulted in improper gel formation. However, recipe GSH₆₀₋₁₈, which had the same monomer concentration but with higher PEG concentration, was suitable for printing hydrogel pillars. Therefore, we conclude that both monomer concentration and the viscosity of the pre-gel solution have to be considered during development. Figure 5.1 shows the TPP printed 3D structure using GSH₇₀₋₁₈. Both the pillars and the U structures were printed with resolutions of order of a few microns. All the rest of the recipes, except for GSH₆₀₋₉ and GSH₇₅₋₁₅, were suitable for printing GSH pillars, and these pillars were then exposed to glucose solutions. The summary that includes pillar size changes in response to glucose among all printable recipes with associated pre-gel viscosities is shown in Figure 5.2. As expected, all the pre-gel solutions exhibit higher viscosity values than a typical dilute aqueous solution because of the PEG dosage.

In comparison to the results in Figure 4.3 (c) in Chapter 4, the pillar shrinking percentage in this study are all smaller than the standard recipe GSH pillars, by least a factor of three. Adding PEG to the pre-gel solution not only boosts up the viscosity but can also increase the shear modulus of GSH. According to Flory's theory, GSH volume change can be expressed as the summation of a few osmotic contributions [14], one of which is the shear modulus, which represents the entropic penalty associated with polymer chain stretching. As a result, a GSH might be expected to swell less when it has a higher shear modulus. Figure 5.3 (a) demonstrates that the shrinking percentage decreases as the

viscosity increases. Figure 5.3 (b) also confirms the same reasoning that viscosity enhancement can compromise the amount of shrinking in hydrogels. However, GSH₆₅₋₁₀, GSH₆₅₋₁₄, GSH₆₅₋₁₈ don't seem to follow the trend. We suspect other osmotic contributions might dominate for GSHs with 65 wt% monomer concentration.

5.3.2 In-Situ TPP GSH as a Flow Guide in a Microchannel

Based on the shrinking result of GSH in Figure 5.2, we here chose to print GSH₇₀₋₁₅ in a microfluidic channel in-situ as a flow guide. Figure 5.4 (a) shows a scheme of the predicted flow pattern while deionized (DI) water flows into the microchannel. The channel was first filled with 0.05 M PBS ($t = 0$ min). At $t = 0.5$ min, the bottom stream was blocked and the reservoir connected to the top channel was changed to DI water. At $t = 1$ min, DI water successfully reached the microchannel and should cause the GSH to swell, thereby not allowing DI water to go through and reach the bottom stream. The experiment result follows our expectation as indicated from Figure 5.4 (b) to Figure 5.4 (i).

Figure 5.5 (a) shows a scheme of the predicted flow pattern while 10 mM glucose flows into the microchannel. The channel was first filled with 0.05 M PBS ($t = 0$ min). At $t = 0.5$ min, the bottom stream was blocked and the reservoir connected to the top channel was changed to the solution that contains 10 mM glucose in 0.05 M PBS. At $t = 1$ min, the glucose solution successfully reached the microchannel and should result in shrinking the GSH, thereby allowing the two channels to be interconnected. The glucose solution was then gradually withdrawn more and more to the bottom channel. The experiment result follows our expectation as indicated from Figure 5.5 (b) to Figure 5.5 (i). After 2.5 min of glucose solution introduced into the top channel, we can visualize solution being

withdrawn from the top channel to the bottom channel (Figure 5.5 (f)). The result here shows that we are able to use glucose and DI water as stimuli to direct the flow in the microchannel.

5.4 Conclusions

In this work, we have developed the pre-gel solution of glucose-sensitive hydrogels that can be used in Nanoscribe to do 3D lithography. The preparation of the pre-gel solution was designed in a 4x3 matrix, which includes four different monomer concentrations by three different PEG concentrations. We found that the printability depended both on monomer concentration and viscosity. Two printed structures, University of Utah Logo and micro-pillars, appeared in high resolutions investigated under the optical microscope. TPP-printed GSH pillars were able to shrink in response to glucose. However, the amount was significantly less than the original recipe, which had lower monomer concentration and no PEG. The viscosity of the pre-gel solution was negatively correlated to the pillar shrinking ratio with exceptions of GSH₆₅₋₁₄ and GSH₆₅₋₁₈. Lastly, we fabricated a microfluidic device that contained TPP-printed GSH as a valve to control the flow between the two microchannels. We successfully guided the flow from one channel to the other through introducing glucose, and the response time was of order 3 minutes.

5.5 References

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Table 5.1
Concentration of monomers and poly (ethylene glycol) in the various
glucose sensitive hydrogels studied in the current work

	Monomer Conc. (wt%)	PEG Conc. (wt%)
GSH ₆₀₋₉	60%	9.0%
GSH ₆₀₋₁₈	60%	18%
GSH ₆₅₋₁₀	65%	9.6%
GSH ₆₅₋₁₄	65%	14%
GSH ₆₅₋₁₈	65%	18%
GSH ₇₀₋₁₀	70%	10%
GSH ₇₀₋₁₅	70%	15%
GSH ₇₀₋₁₈	70%	18%
GSH ₇₅₋₁₁	75%	11%
GSH ₇₅₋₁₅	75%	15%

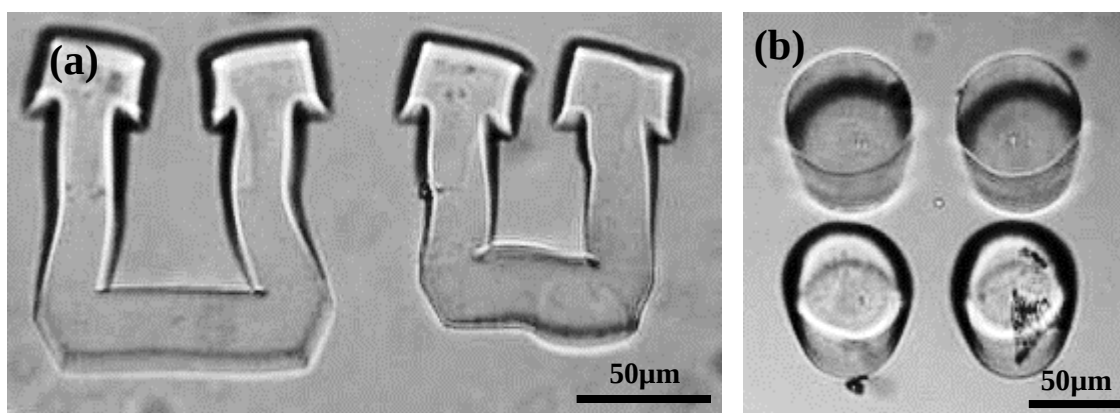


Figure 5.1 3D structures of glucose-sensitive hydrogel printed using two-photon polymerization.

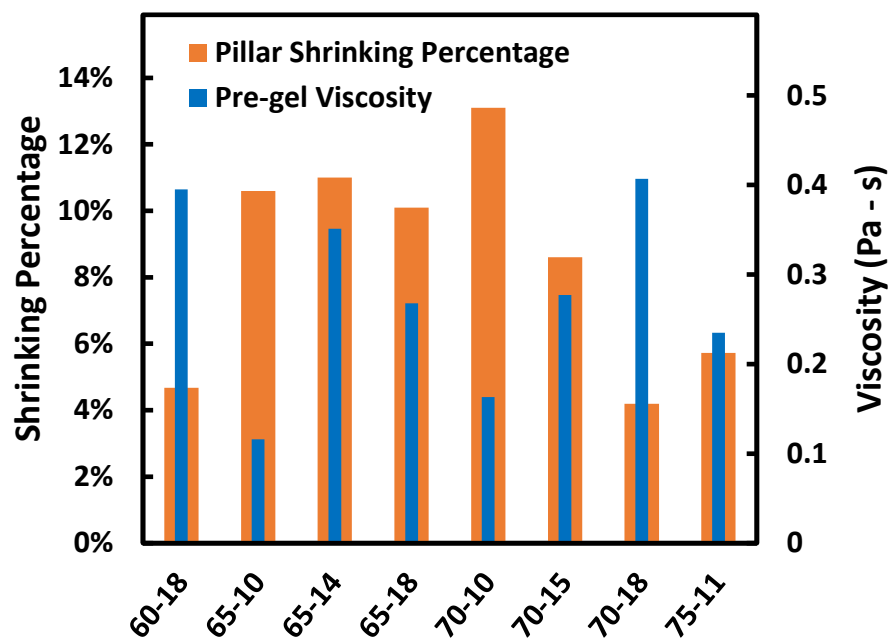


Figure 5.2 Pillar shrinking percentage with respect to viscosity of the pre-gel solution for the overall comparison. The pillar shrinking percentage was calculated using the change of the pillar top-view area from 0 mM to 10 mM

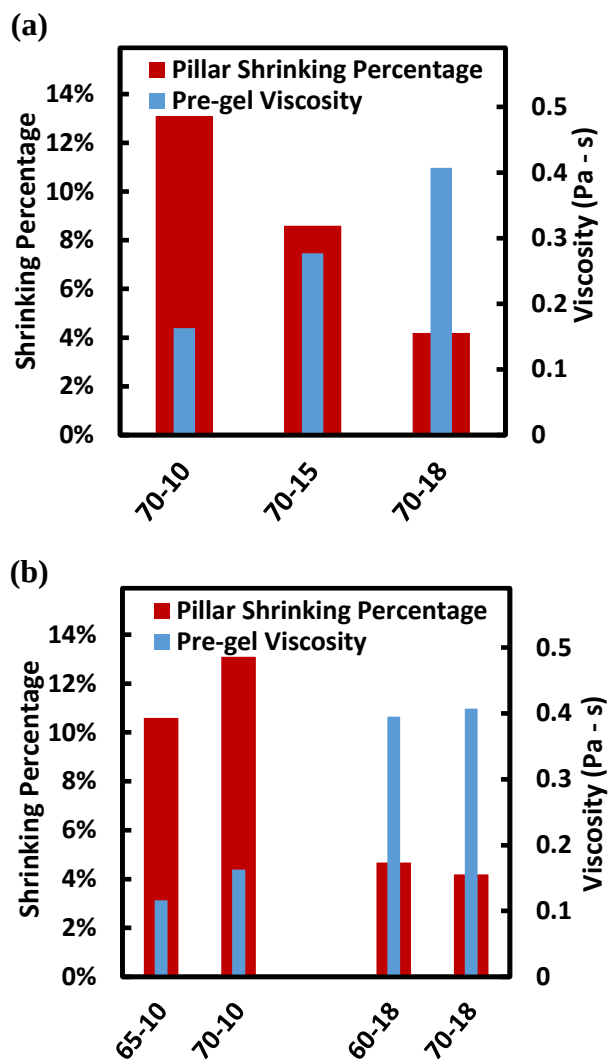


Figure 5.3 Pillar shrinking percentage with respect to viscosity of pre-gel solution: comparison between the fixed monomer concentration and the fixed polyethylene glycol concentration. The pillar shrinking percentage was calculated using the change of the pillar top-view area from 0 mM to 10 mM glucose under an optical microscope.

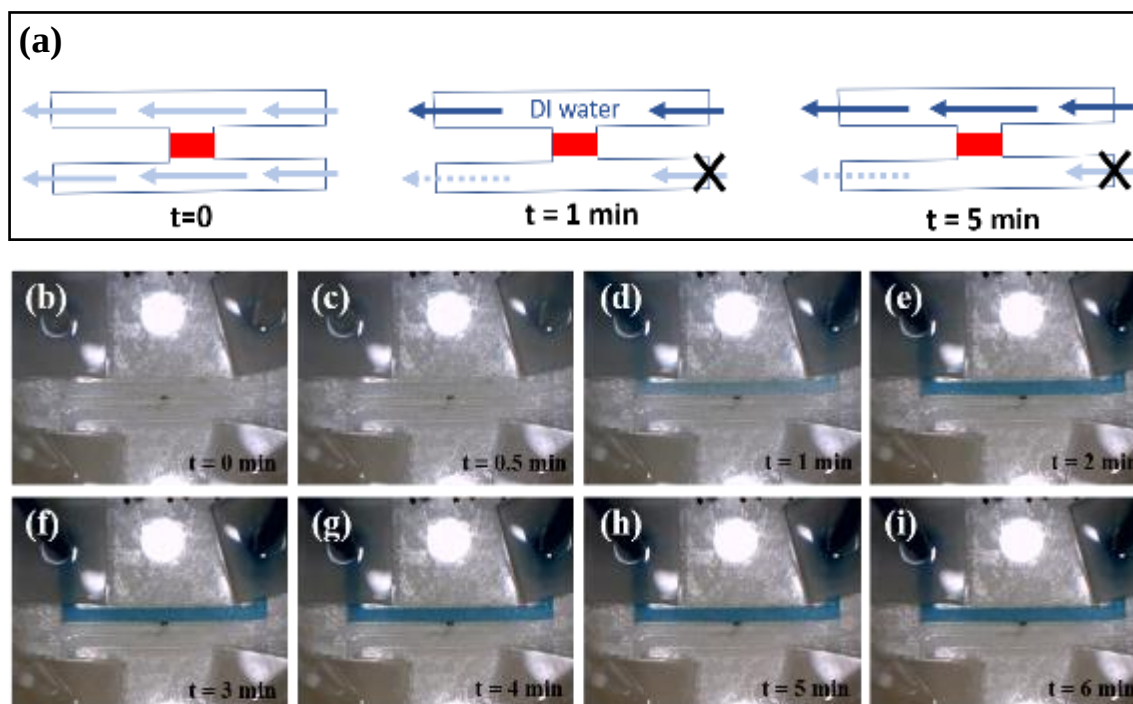


Figure 5.4. Two photon polymerized hydrogel behavior test in DI water. Figure (a) demonstrates a scheme that predicts the flow pattern in the microchannel with DI water introduced to the top channel. The arrows with solid lines in light blue color indicate flow that contains 0.05 M PBS. The dash line represents the stagnant flow. The red rectangle that locates between the two channels represents the glucose-sensitive hydrogel (GSH), which behaves as the fluid flow guide. Figures (b), (c), (d), (e), (f), (g), (h), and (i) show the top-view of the microchannel at $t = 0$ min, 0.5 min, 1 min, 2 min, 3 min, 4 min, 5 min, and 6 min, respectively. DI water, added with blue food coloring for visualization, first reached the top channel at $t = 1$ min. DI water swells the GSH, thereby blocking the flow path between the upper and lower channels.

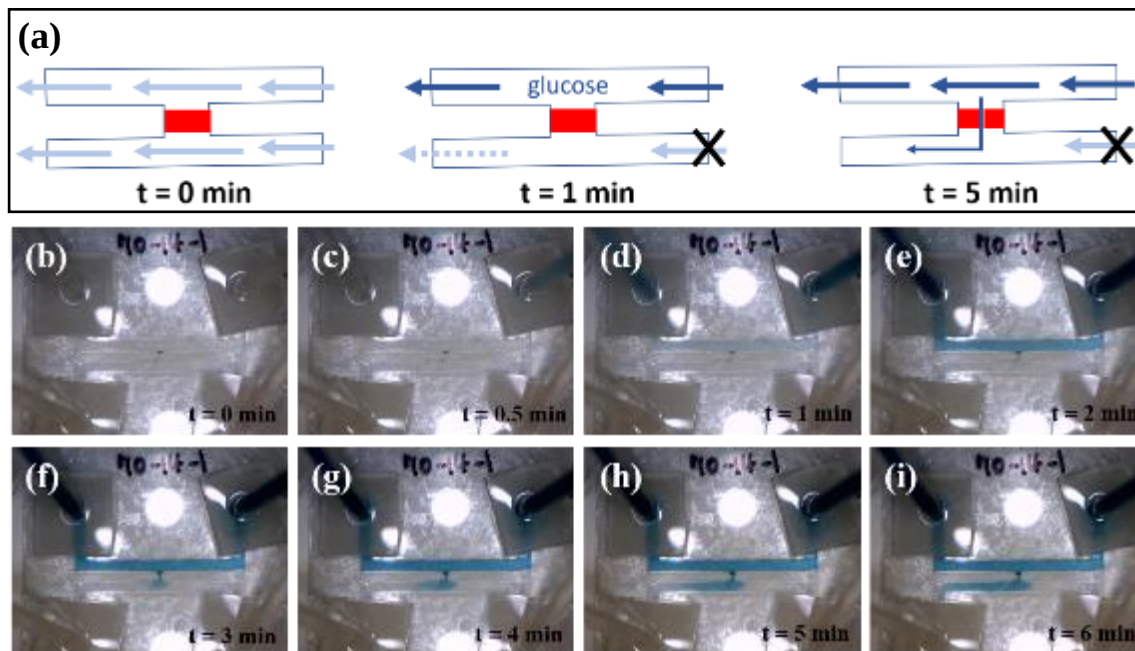


Figure 5.5 Two photon polymerized hydrogel behavior test in glucose. Figure (a) shows the scheme that predicts the flow pattern in the microchannel with 10 mM glucose in 0.05 M PBS introduced to the top channel. The arrows with solid lines in light blue color indicate flow that contains only 0.05 M PBS without glucose. The dash line represents the stagnant flow. The red rectangle that locates between the two channels represents the glucose-sensitive hydrogel (GSH), which behaves as the fluid flow guide. Figures (b), (c), (d), (e), (f), (g), (h), and (i) show the top-view of the microchannel at $t = 0$ min, 0.5 min, 1 min, 2 min, 3 min, 4 min, 5 min, and 6 min, respectively. 10 mM glucose solution was added with food coloring (blue) for visualization. Introduction of glucose shrinks the GSH valve and allows flow to enter the bottom channel from the top channel.

CHAPTER 6

CONCLUSIONS AND FUTURE DIRECTIONS

6.1 Conclusions

The goal of this work was to develop a new type of smart-hydrogel-based microfluidic sensor that can substantially improve the response time of smart hydrogel sensors for Point-of-Care applications. The development firstly includes sensor fabrication with the evaluation of the new transduction technique “Resistive Channel Sensing (RCS)” among different electrode and channel designs. Secondly, the sensor was integrated with smart hydrogels to continuously monitor the volume change of smart hydrogels in response to the environmental stimuli. Results of the research support the following conclusions:

1. For the proof-of-principle of RCS, we found that a microsensor integrated with PBS-filled microchannels and electrodes can successfully transduce the microchannel resistance using a potentiostat. The results obtained by a potentiostat with four-electrode configuration show better proportionality of the resistance of the microchannel to the resistivity of PBS than using a potentiostat with three-electrode configuration. Having a side channel to keep the reference electrode at a constant absolute potential can help avoid the corrosion of the electrodes.
2. Arrays of pH-sensitive smart hydrogel pillars were successfully fabricated in a microfluidic channel in-situ using a low-cost photolithography approach that

includes a darkfield mask and a collimated light source in the cleanroom. Such a microfluidic pH sensor obtained reversible and reproducible electronic signals. Moreover, the response time of the microfluidic pH sensor was shown to decrease through downsizing the diameter of the hydrogel pillars.

3. An improved method with no need for cleanroom facilities was able to fabricate arrays of glucose-sensitive smart hydrogel pillars in a microfluidic channel using the similar inexpensive photolithography approach. The response time of such microfluidic glucose sensor was of order 5 minutes obtained using potentiostat transduction measurements and was faster than microscope observations of pillar diameter. This response time improvement can be attributed to the ionic exchange between the pillars and the surrounding solution that occurs when the pillars change their net charge.
4. To further improve the response time of smart hydrogels through downsizing, we aim to develop pre-gel solutions that can be used in Nanoscribe to do 3D lithography. By modifying the current glucose-sensitive hydrogels recipe by boosting both monomer concentration and viscosity, we successfully printed structures with high resolution including arrays of smart hydrogel pillars in a small drop of the pre-gel solution. The pillars showed volume change in response to glucose, but the magnitude was smaller by at least a factor of three than the conventional formulations. Moreover, we were able to print glucose-sensitive hydrogel valves in-situ in a microfluidic channel. Through introducing glucose, the printed gel was able to behave like a smart valve to guide the fluid flow between two microfluidics channels

6.2 Future Work

Based on the work accomplished in this dissertation, the sensing platform investigated, resistive channel sensing (RCS), shows its capability in monitoring the resistance changes of a microchannel that occur when filled with solutions of different conductivities. This sensing approach was successfully applied to obtain reversible smart-hydrogel-based pH and glucose sensors. However, the sensors suffer from a baseline drift that may be attribute to electrode instability. For example, the gold-electrode sensors used were found to be corroded after 2 weeks of daily heavy usage. Given that the glucose sensors require a few minutes, the shelf-life of the electrodes might be prolonged if one uses the sensor only when needed for promptly measurements. However, there could be much effort spent on these aspects of material science for sensor stability improvement. One potential approach to resolve such issues in the future would be to modify the electrode by changing the electrode materials or by better characterizing the surfaces of the current material.

For hydrogel response time improvement, two-photon polymerization (TPP) was found to be capable of printing glucose-sensitive hydrogels as small as a few microns with high resolution. Colorimetric results from a microfluidic chip have demonstrated that a printed hydrogel valve was printed in-situ and could guide the flow of order a few minutes after reaching glucose. Unfortunately, in order to use TPP, the monomer concentration and viscosity need to be substantially increased, which reduces the magnitude of smart hydrogel response. In the future, additional factors could be considered to optimize hydrogel response time, as well as reduce the response time. It will be desirable to in-situ fabricate smart hydrogels structures with even higher surface area-to-volume ratio in an

RCS microfluidic sensor chip using TPP.

The smart-hydrogel-based glucose and pH sensors developed in this study successfully transduced hydrogel volume changes to electrical current signals using the RCS approach applied to environmental pH and glucose sensing. However, stronger interest in the current point-of-care sensor markets is focused on detecting environmental contaminants (e.g., heavy metals) and virus diagnosis (e.g., Zika) [1]. To make progress in these newer sensing areas, the development of more advanced stimuli-responsive hydrogels such as aptamer-based hydrogels is needed. Unfortunately, due to DNA damage, the synthesis of smart hydrogels that contain DNA aptamers has thus far been restricted from the use of photolithography, which is one of the more promising methods for miniaturization. In order to overcome this limitation, in the future one can focus on DNA aptamer-based hydrogel fabrication using darkfield mask photolithography with the collimated light source that has a bandwidth >400 nm to potentially reduce the DNA damage. Alternatively, one can focus on further chemical development for the DNA aptamer-based pre-gel solution to make them more suitable for TPP fabrication. Research has shown that intense, ultrashort, near-infrared laser pulses may damage DNA but are intensity-dependent [2]. Therefore, finding ways that can evaluate the integrity of DNA aptamers after photolithography will be equally important as monitoring the gel swelling behavior.

Lastly, bring the sensor from the lab to the point-of-care sensor market, the sensor needs a comprehensive report provided to government agencies, for example U.S. Food and Drug Administration. Firstly, the report needs to include bench performance data and clinical performance data. Therefore, the report needs to have results for regular

performance tests as well as a series of failure tests to foresee what factors can potentially cause sensor malfunction. Such factors might include stimuli cross-interference in hydrogel response, potential unwanted redox reactions that might occur at the electrodes owing to different carrying fluids, and the stability and the reproducibility among different environmental conditions such as temperature and humidity. Secondly, biocompatibility, sterility and self-life testing reports are also required and have to follow certain standardized methods such as International Organization for Standardization (ISO) and the American Society for Testing and Materials (ASTM).

In summary, future work includes further improvement of electrode properties, smart hydrogel development that can potentially improve the response time, and a comprehensive report to satisfy government regulations so that the sensor can move from the lab to the market.

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